

- Q21. In Soft Prompt Tuning, prompt vectors are inserted into the model at which stage of processing?
- A) After the transformer layers, just before the output head
 - B) Inside every attention block to modify query-key interactions
 - C) Before the transformer layers, concatenated to the input token embeddings
 - D) Replacing positional encodings inside the embedding matrix
- Q22. How does P-Tuning improve optimization stability?
- A. By learning continuous prompt embeddings through a small neural network
 - B. By restricting prompts to the fixed pre-trained vocabulary
 - C. By forcing all prompts to be discrete tokens
 - D. By removing positional encodings from prompt tokens
- Q23. In soft prompt tuning, which parameter set is optimized during training?
- A. Only the model weights θ , because prompts embeddings are frozen and act as fixed adapters.
 - B. Both the prompt vectors P and the model weights θ .
 - C. Neither the prompts nor the model weights, Only final classification layer
 - D. Only the prompt vectors P , because the pre-trained model is frozen, reducing trainable parameters
- Q24. Which method can potentially approach full fine-tuning performance with fewer parameters?
- A. VPT-Shallow
 - B. VPT-Deep
 - C. Zero-shot prompting
 - D. CNN fine-tuning
- Q25. If P-Tuning uses an MLP with hidden size h , and initial prompt size is $p \times d$, then number of MLP parameters is approximately:
- A. $p+d$
 - B. pd
 - C. $d \times h + h \times p$
 - D. $p \times h \times d$
-



राजीव गांधी पेट्रोलियम प्रौद्योगिकी संस्थान, बंगलुरु परिसर
Rajiv Gandhi Institute of Petroleum Technology, Bengaluru Campus

(राष्ट्रीय महत्व का संस्थान)
Institution of National Importance

कंबलिपुर, सुलेबेल हिल, होसकोटे, कर्नाटक - 562165, भारत
Kambalipura, Sulibele Hill, Hoskote, Bengaluru, Karnataka - 562165, India

Roll no. 22 43002

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B.Tech IDD-22 Fourth Year Even Semester
End-Semester Examination 2025-26
Subject: MT5213 Sustainability and Climate Change

Time: 2.5 hrs.

Full Marks: 50

Note: Attempt all the questions. All questions carry equal marks
The use of a scientific calculator is allowed

1. A Define Sustainable Development and explain the concept of Triple Bottom Line **5M**
B What are the Sustainable Development Goals (SDGs)? Explain their relevance to climate change. **5M**
2. A Explain the business case for sustainability. How can sustainability create competitive advantage and long-term value for businesses? **5M**
B What is Life Cycle Assessment (LCA)? Explain its importance in sustainable supply chain management. **5M**
3. A What is sustainable marketing and branding? Explain how businesses can communicate sustainability effectively to consumers. **5M**
B Explain the importance of measuring and reporting sustainability performance. What are the key frameworks and metrics used? **5M**
4. A What is change management for sustainability? Explain the barriers organizations face while implementing sustainability initiatives. **5M**
B Why are collaboration and leadership important for achieving sustainability goals in organizations? **5M**
5. Case Scenario Analysis **10M**
a. A company consumes 50,000 kWh of electricity per year. After installing energy-efficient equipment, electricity consumption reduces to 38,000 kWh per year. If the cost of electricity is ₹8 per kWh, calculate the annual cost savings.



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(राष्ट्रीय महत्व का संस्थान Institution of National Importance)

कंबलिपुरा, सुलिबेल रोड, होस्कोटे, बंगलुरु, कर्नाटक - 562165, भारत Kambalipura, Sulibele Road, Hoskote, Bengaluru, Karnataka - 562165, India

- How does energy efficiency contribute to both cost savings and environmental sustainability?
- What additional measures can the company take to further reduce energy consumption?
- How can these savings improve the company's long-term competitiveness?

b. A company produces 10,000 plastic bottles every month. By adopting recyclable packaging, it reduces plastic use by 25%. If each bottle originally required 20 grams of plastic, calculate the reduction in plastic use per month.

- What are the environmental benefits of reducing plastic consumption?
- How can recyclable packaging improve the company's brand image?
- What challenges might the company face while switching to recyclable materials?

RAJIV GANDHI INSTITUTE OF PETROLEUM TECHNOLOGY, JAIS AMETHI
B.Tech IDD 5th YEAR

MID-SEM EXAMINATION, (Jan - May 2026)

INSTRUCTOR: Dr. PALLABI SAIKIA

SUBJECT: Digital Image Processing
TIME: 2 Hrs

COURSE NUMBER: CS431
FULL MARKS: 30

INSTRUCTIONS:

1. Do not write anything other than your roll number on the question paper.
2. Assume suitable data wherever essential and mention it clearly.

Q1. (5 Marks) Consider the following binary image (object pixels = 1):

```

1 0 1 1 0
1 1 0 1 0
0 1 1 0 0
1 0 1 1 1
0 0 0 1 0

```

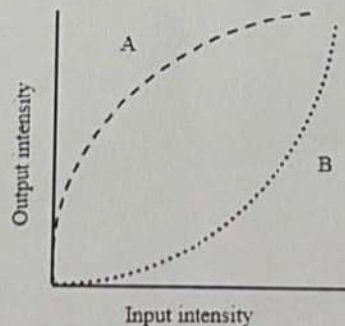
For $V=\{1\}$:

- A. Determine the number of connected components using 4-connectivity.
- B. Determine the number of connected components using 8-connectivity.
- C. Determine the number of connected components using m-connectivity.
- D. Find the shortest path between pixels at locations (1,1) and (4,5) adjacent under 4 connectivity and m-connectivity? Justify your answer.

Q2. (3 Marks) A pixel in an RGB image has the values: $R=120, G=200, B=80$ (Assume intensity range: 0–255)

- A. Compute the luminance and chrominance components of the image
- B. Identify the approximate Hue category in the HSV model based on the dominant channel. Justify your answer.

Q3. (2 Marks) Two gray-level transformation curves A and B are shown below.



Answer the following:

- I. Identify the type of transformation represented by curve A and curve B.
- II. Which curve is suitable for enhancing a dark image and which is suitable for enhancing a bright image? Justify your answer.
- III. Write the mathematical expression of the general power-law (gamma) transformation and explain the role of the parameter γ . (Assume gray levels are in the range 0 to $L-1$)

Part B: Financial Evaluation

5. Calculate the annual revenue from hydrogen sales (₹/year).
6. Determine the annual net cash flow (₹/year).
7. Compute the Net Present Value (NPV) of the project over its lifetime.

Part C: Levelised Cost of Hydrogen (LCOH)

8. Using discounted cash flow principles, calculate the Levelised Cost of Hydrogen (LCOH) in ₹ per kg of H₂.

Part D: Policy Interpretation

9. Compare the calculated LCOH with the NGHM benchmark of ₹300/kg H₂ and comment on:
 - Whether the project is financially competitive without policy support
 - The likely role of capital subsidies, concessional power tariffs, or PLI incentives

Caselet-2: Life Cycle Assessment using ReCiPe 2016

InfByte Technologies Ltd. is an Indian technology company that designs and manufactures a smart wearable fitness device used to monitor physical activity and health parameters.

The device contains:

- A lithium-ion battery
- Semiconductor chips and sensors
- Plastic casing
- Wireless connectivity
- Requires regular charging using grid electricity

The company intends to evaluate the environmental sustainability of this product using a Life Cycle Assessment (LCA) approach and has selected the ReCiPe 2016 impact assessment method.

The assessment considers the entire life cycle of the product, including:

- Raw material extraction
- Component manufacturing
- Product assembly
- Use phase (electricity for charging)
- End-of-life disposal or recycling

(a) Using the context above, identify any four life cycle inventory flows associated with the smart wearable device. Briefly explain why each flow is environmentally relevant.

(b) Referring to the ReCiPe 2016 framework:

1. Map any two of the identified inventory flows to appropriate midpoint impact categories (e.g., Climate Change, Human Toxicity, Mineral Resource Scarcity).
2. Briefly justify your mapping.

(c) Endpoint Damage Categories

Explain how the selected midpoint impact categories can lead to damage in the following endpoint categories

- Human Health
- Resource Availability

Your answer should clearly describe the cause-effect relationship.

(d) Design or Usage Improvement

Suggest two improvements (design or usage related) that could reduce the environmental impacts of the smart wearable device.

For each improvement, clearly mention:

- one midpoint category, and
- one endpoint category that would be positively affected.



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बंगलुरु परिसर **Bengaluru Campus**
(राष्ट्रीय महत्व का संस्थान **Institution of National Importance**)

कंबलिपुरा, सुलीबेल हिल, होसकोटे, बंगलुरु, कर्नाटक - 562165, भारत **Kambalipura, Sulibele Hill, Hoskote, Bengaluru, Karnataka - 562165, India**

Although the country is a signatory to the Paris Agreement, it continues to rely heavily on coal for electricity due to energy security concerns and industrial growth targets.

The government must now balance economic development, environmental sustainability, and social welfare.

Question:

- (a) Analyze how climate change impacts create interlinkages between at least three sectors. (3 Marks)
- (b) Evaluate the policy trade-off between economic growth and emission reduction. (3 Marks)
- (c) Propose two mitigation and two adaptation strategies with justification. (2 Marks)
- (d) Critically comment on the role of international cooperation in addressing climate change. (2 Marks)

Time 3 hrs

Full Marks: 50

✓ Qu1. How do the Big Five personality traits influence workplace behavior and performance? Can certain traits predict success in specific job roles? Substantiate with examples. (10)

✓ Qu2. How has the digital age transformed traditional grapevine communication into online informal networks (e.g., WhatsApp, Slack, social media), and what new challenges does this pose? (10)

✓ Qu3. Please read the following case study and respond to the questions that follow.

"Remote Week" at HubSpot

When Rachel Leist was working in the marketing team at HubSpot, a technology company based in Cambridge, Massachusetts, the company noticed a change as it continued to grow. Many teams were now located in different parts of the world, which meant that meetings were often held online. Rachel's marketing team wanted to improve how they worked together and also help the company learn from their experience. As a result, they introduced a special initiative called "Remote Week."

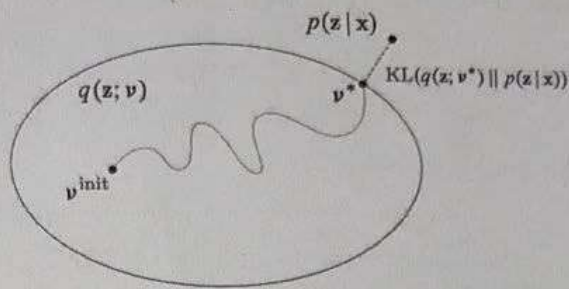
During Remote Week, the team decided to conduct all their meetings through video calls. They experimented with different types of meetings, including one-to-one meetings, cross-functional meetings, and meetings involving the entire team. At the end of the week, feedback was collected from all team members to understand what worked well and what did not.

One important finding was that having a facilitator during video meetings made a significant difference. The facilitator ensured that everyone had an opportunity to speak. The team found that sometimes facilitators would see people making "goldfish faces" (opening and closing their mouths without saying anything as dominant viewpoints prevailed). The facilitator helped prevent this by managing turn-taking so that all voices were heard. Facilitators also paid attention to side conversations and encouraged participants to share those points with the entire group. Additionally, having everyone join the meeting from separate rooms helped reduce private side discussions.

The team also realized that informal, spontaneous conversations are very useful but difficult to achieve in virtual settings. To solve this, they created alternatives. Team members agreed to quickly join short video calls whenever someone had a question. If a person was unavailable,

- **Hardware/Prototype** (EV sub-assembly, sensor board, embedded controller, test jig)
- **Analytics/Optimization** (fault prediction, yield analysis, route/energy optimization) **(1x8)**

1.	Some of the important parts of the role JD are problem-solving, working with others, managing tools and versions, documentation, talking to people who aren't technical, measuring and tracking progress, and delivering in small steps (Agile/PDCA). - Pick five important keywords from the JD, such as problem-solving, working together, documentation, metrics/KPIs, iterative delivery/Agile/PDCA, and tooling/version control. Connect each one to a part of the project you chose. - Use the template below to turn one of your project's contributions into a quantified résumé bullet: Action verb + task + tool/tech/process + metric (impact on time/cost/quality/safety/reliability/energy). - Write a 120–140-word email to a hiring manager that - References two or three JD keywords, - Gives a brief overview of the effects of your project using numbers, and - States when they are available for an interview and when they can start. - Make a 30–60–90 day plan (two bullets for each phase) for joining a cross-functional team or lab: 30 days: setting up the environment, writing documentation, shadowing, and doing a small fix or replication task. 60 days: own a scoped feature, test and validate it, and show it to stakeholders once a week. 90 days: performance hardening or scaling up, handing over documentation, making a risk register, and making a proposal for how to improve next quarter.	2+2+2+2
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- A. Explain why the true posterior $p(z|x)$ generally lies outside the variational family $q(z;v)$
- B. Show how minimizing $KL(q(z;v) || p(z|x))$ is equivalent to maximizing the Evidence Lower Bound (ELBO).
- C. What does the path from v_{init} to v^* represent? Interpret the role of v_{init} and v^* in the optimization procedure.

Q3. Explain the complete training and sampling pipeline of a diffusion model. A Denoising [10] Diffusion Probabilistic Model (DDPM) is defined by a forward noising process:

$$q(x_t|x_{t-1}) = N(x_t; \sqrt{1-\beta_t} x_{t-1}, \beta_t I).$$

- A. What is β_t and its significance in the diffusion model
- B. Formulate equation of x_t as markov process. Why does x_T converge to Gaussian noise as $t \rightarrow T$
- C. Write the mathematical representation of the generative process $p_\theta(x_{t-1}|x_t)$ in the diffusion model and how we can relate it with the variational inference of VAE?
- D. Represent both the Forward and Reverse Diffusion Process in Stochastic Differential Equation

OR

Q3. Diffusion models, while highly effective for generative tasks, encounter two key challenges: [10] slow sampling and scalability to high-resolution (conditional) generation. State and explain different strategies adopted in diffusion models with respect to the following :

- A. Discuss the strategies employed to accelerate the sampling process in diffusion models. Explain their underlying working principles and architectural approaches wherever applicable.
- B. Explain how diffusion models can be adapted for high-resolution conditional generation. Illustrate the architectural techniques and guidance mechanisms used.

Roll No. _____

Rajiv Gandhi Institute of Petroleum Technology

(An Institute of National Importance)

Mid-Semester Examination, Odd Semester, 2025-2026

B. Tech. 4th YEAR (CSE + IT + CSD + MNC + EC) + IDD (4th + 5th) YEAR

Course Name : Computer Vision and Pattern Recognition
Course Code : CS454

Time: 2 Hours
Max. Marks: 30

Instructions:

1. Do not write anything other than your roll number on the question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Q1: Explain the concept of local connectivity in Convolutional Neural Networks (CNNs) with an [10] example. Why is it preferred over full connectivity in image-based tasks? Consider a grayscale input image of size 32×32 , which is passed through a convolutional layer with the following configuration:

- Filter/kernel size: 5×5
- Number of filters: 6
- Stride: 1
- Padding: 0
- Each filter connects only locally and uses weight sharing.

Answer the following:

- A. What is the size (height $\times$ width) of the output feature map for each filter?
- B. How many total connections (weights) would have been needed if this was a fully connected layer with the same number of outputs per filter (no weight sharing)?
- C. How many total learnable parameters (weights + biases) are present in this convolutional layer?

Note: Assume parameters if required

OR

Q1. Discuss in detail design principles and working architectures of any two following models:

- A. VGGNet for ImageNet Image Classification
- B. ResNet for ImageNet Image Classification
- C. HOG Descriptors in Deformable Part Model for Object Detection
- D. R-CNN for object detection.

[10]

Q2: Mention any three Attribute Selection Measures used in popular Decision Tree algorithms [10] and write their mathematical expressions. Suppose you are tasked with predicting the energy consumption behavior of households based on three categorical variables: Region (North, South, East, West), Type of Residence (Apartment, House, Condo), and Primary Heating Source (Electricity, Gas, Oil). Construct a decision tree to predict the energy consumption behavior based on these variables using any of the attribute selection criteria.

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B. Tech./IDD End Semester Examination (Winter Sem, 2025-26)

Subject: Sustainability & Climate Change

Time: 3hrs

Full Marks: 60

PART-A (45 marks): Q1–20 marks; Q2 to Q6, 5 marks each -- to be answered in the main answer script

PART-B (15 marks): Indicate the options in the answer boxes below each question; and ensure that the question paper is submitted along with the main answer script, failing which Part B will not be evaluated.

Caselet-1: Green Hydrogen Electrolyzer Project under India's National Green Hydrogen Mission

Aarohan Green Hydrogen Private Limited (AGHPL) is an Indian clean-energy company planning to set up a renewable electricity-based green hydrogen production facility in western India. The project is proposed under India's National Green Hydrogen Mission (NGHM) and aims to supply green hydrogen to nearby fertilizer and refinery units currently dependent on grey hydrogen.

The company plans to install a grid-connected electrolyzer system powered by renewable electricity procured through long-term power purchase agreements (PPAs).

Technical Clarification (for Students):

The electrolyzer capacity (MW) represents the maximum electrical power input that the electrolyzer can draw at full load.

Hydrogen production is related to electrolyzer capacity using the specific electricity consumption as follows:

$$\text{Hydrogen production (kg/hour)} = \frac{\text{Electrolyzer capacity (kW)}}{\text{Electricity consumption (kWh/kg H}_2\text{)}}$$

Annual hydrogen production can be obtained by multiplying hourly production by annual operating hours.

Project and Financial Data

Parameter	Value
Electrolyzer capacity	20 MW
Project lifetime	20 years
Annual operating hours	6,000 hours
Electricity consumption	52 kWh/kg H ₂
Capital investment (Year 0)	₹ 2,520 crore
Decommissioning cost (Year 20)	₹ 180 crore
Renewable electricity tariff	₹ 4.05 per kWh
O&M cost	₹ 54 per kg H ₂
Hydrogen selling price	₹ 405 per kg
Discount rate	8% per annum

IM = 103 kW

LCOH

Policy Context: National Green Hydrogen Mission (NGHM)

Under the National Green Hydrogen Mission, the Government of India has indicated an aspirational cost target of approximately ₹300 per kg of green hydrogen to make it competitive with fossil-based hydrogen. Projects achieving or approaching this cost are considered strategically important for:

- Import substitution
- Industrial decarbonisation
- Long-term export competitiveness

Answer the following questions:

Part A: Production and Cost Estimation

1. Calculate the annual hydrogen production of the electrolyzer plant (kg/year).
2. Estimate the annual electricity cost (₹/year).
3. Estimate the annual operation and maintenance (O&M) cost (₹/year).
4. Calculate the total annual operating cost (₹/year).

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B. Tech./IDD End Semester Examination (Winter Sem, 2025-26)

Subject: Sustainability & Climate Change

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4. Calculate the total annual operating cost (₹/year).

Time 2 hrs

Full Marks: 30

1. What is the difference between Energy Transition and Just Energy Transition. Elucidate with suitable examples. (7)
2. How can Michael Cernea's Impoverishment Risks and Reconstruction (IRR) model be applied to manage risks in communities affected by the implementation of renewable energy projects? (8)
3. Read the case study carefully and answer the questions that follow:

Case Study: Public Resistance to Wind Farms in the UK

The United Kingdom has positioned itself as a leader in renewable energy, particularly in offshore and onshore wind. However, efforts to expand onshore wind farms have repeatedly encountered strong local opposition. Surveys by the UK government's Public Attitudes Tracker (Spring 2025) reveal that while around 85–90% of people nationally support renewable energy, a much smaller proportion are comfortable with wind projects located in their own local area. This disjuncture, often captured under the phrase "Not in My Backyard" (NIMBY), highlights the gap between broad national support for clean energy and opposition to its implementation in specific places.)

At its core, NIMBYism reflects a tension between collective environmental benefits and individual or community concerns. In places such as Lincolnshire, residents like Jane and Julian Davis famously challenged a local wind farm in court, claiming that the low-frequency noise from turbines disrupted their sleep and health, eventually forcing them to abandon their home (BBC, 2011). In Scotland's Cabrach region, where over 70 turbines are already operating and more are planned, community groups argue that the cumulative expansion has reached "saturation point," threatening rural character, heritage landscapes, and local identity (STV News, 2024). Such objections often focus on aesthetic disruption — wind turbines are seen as intrusions into cherished rural scenery, symbols of industrialization that clash with cultural identity. Others cite noise pollution, declining property values, or disruption to daily life. These concerns demonstrate that opposition is not merely economic but rooted in deep attachments to place and lifestyle.)

P.T.O



Rajiv Gandhi Institute of Petroleum Technology
Bengaluru Campus
Mid-Semester Examination
(ODD Semester, 2025-26)

Max. Marks: 30 **Duration:** 120 Min. **Date:** 26/9/2025 **Roll No:**
Programme/Semester: B. Tech 7th Semester **Time:** 03:00 PM-05:00 PM
Course Title: Employability Skills and Industry Readiness

General Instructions

1. Try to answer all of the questions.
2. Use the context of your own field (software, IT, hardware, electronics, analytics, EV systems, controls, etc.).
3. Be brief but meaningful.
4. If you can, put numbers on the effects (% , time, cost, reliability, energy, safety, accuracy, throughput).
5. Write in a way that is easy to read. Use bullets whenever it is required.

S.NO	Questions	Marks
SECTION A (Objectives)		
Attempt ALL Parts and select the best option		(10x1)
1.	The STAR method stands for: a) Situation, Task, Action, Result b) Scope, Test, Act, Review c) State, Target, Act, Report d) Source, Tactic, Action, Result	1
2.	A SMART goal is not : a) Specific b) Measurable c) Time-bound d) Vague but inspiring	1
3.	An ATS-friendly résumé is best characterized by: a) Heavy graphics/tables b) Natural use of JD keywords c) Photo-centric layout d) Icons only	1
4.	A strong application email subject is: a) "Job" b) "Application" c) "Applying for Graduate Engineer – Ref: GE-2025 – <Your Name>" d) "Please see"	1
5.	The first step in problem-solving at work is to: a) Pick a favorite tool b) Clearly define the problem with data/evidence c) Escalate d) Form a large committee	1
6.	A good daily stand-up update focuses on: a) Design debates b) What you did, will do, blockers c) Appraisals d) Social news	1
7.	Stakeholder mapping helps you primarily to: a) Avoid bad news b) Identify who is impacted and who decides c) Replace documents d) Skip approvals	1

Interestingly, people often reconcile the contradiction between supporting renewables in principle and opposing them in practice by reframing the debate. Public surveys consistently show respondents stating that they “agree with wind energy” but that “this location is unsuitable” (DESNZ Tracker, 2025). This selective acceptance shows how values and lived experiences shape attitudes toward infrastructure. At the same time, local opposition often overlooks the global dimension of climate change — privileging immediate surroundings over broader planetary needs, a pattern described by critics as a narrowing of perspective. 2

For policymakers and developers, this creates a dilemma: global climate targets demand rapid expansion of wind energy, yet local communities resist when projects come too close. Many planning applications in England have historically been blocked or delayed due to community objections, with The Guardian (2015) reporting that nearly six in ten onshore wind farm applications were rejected at one point. Overcoming these challenges requires more than technical efficiency or financial incentives. Without concrete measures, NIMBY resistance is likely to persist, slowing the pace of energy transition despite widespread recognition of its necessity.

Questions:

- (a) How can the persistence of NIMBY resistance to wind farms be explained through the influence of individual habits and practices, and community routines? (5)
- (b) Why do individuals often support renewable energy in principle but oppose projects near their own homes? What does this reveal about internal contradictions in beliefs and actions? (5)
- (c) What role do civil society organizations, including NGOs, play in shaping community responses to wind farm projects, and how can they help mitigate NIMBY opposition? (5)

Part B: Financial Evaluation

5. Calculate the annual revenue from hydrogen sales (₹/year).
6. Determine the annual net cash flow (₹/year).
7. Compute the Net Present Value (NPV) of the project over its lifetime.

Part C: Levelised Cost of Hydrogen (LCOH)

8. Using discounted cash flow principles, calculate the Levelised Cost of Hydrogen (LCOH) in ₹ per kg of H₂.

Part D: Policy Interpretation

9. Compare the calculated LCOH with the NGHM benchmark of ₹300/kg H₂ and comment on:
 - o Whether the project is financially competitive without policy support
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Caselet-2: Life Cycle Assessment using ReCiPe 2016

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The device contains:

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- Requires regular charging using grid electricity

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The assessment considers the entire life cycle of the product, including:

- Raw material extraction
- Component manufacturing
- Product assembly
- Use phase (electricity for charging)
- End-of-life disposal or recycling

- (a) Using the context above, identify any four life cycle inventory flows associated with the smart wearable device. Briefly explain why each flow is environmentally relevant.
- (b) Referring to the ReCiPe 2016 framework:
 1. Map any two of the identified inventory flows to appropriate midpoint impact categories (e.g., Climate Change, Human Toxicity, Mineral Resource Scarcity).
 2. Briefly justify your mapping.

(c) Endpoint Damage Categories

Explain how the selected midpoint impact categories can lead to damage in the following endpoint categories

- Human Health
- Resource Availability

Your answer should clearly describe the cause-effect relationship.

(d) Design or Usage Improvement

Suggest two improvements (design or usage related) that could reduce the environmental impacts of the smart wearable device.

For each improvement, clearly mention:

- one midpoint category, and
- one endpoint category that would be positively affected.

Example	Region	Type of Residence	Primary Heating Source	Energy Consumption
1	North	House	Gas	High
2	South	Apartment	Electricity	Low
3	East	Condo	Oil	Medium
4	West	House	Gas	High
5	North	House	Electricity	Medium
6	South	Condo	Gas	High
7	East	House	Electricity	Low
8	West	Apartment	Oil	Medium
9	North	Condo	Gas	Low
10	South	House	Oil	High
11	East	Apartment	Electricity	Low
12	West	Condo	Gas	Medium
13	North	Apartment	Oil	High
14	South	House	Gas	Low
15	East	Condo	Electricity	Medium

OR

Q2. What are the assumptions made in the Naïve Bayes algorithm? Suppose we are using the [10] Naive Bayes algorithm to predict the likelihood of an electric vehicle (EV) being purchased based on three categorical variables: Type of EV (Compact, Sedan, SUV), Charging Infrastructure (Home Charging, Public Charging, Workplace Charging), and Range (Short Range, Medium Range, Long Range). Given the following dataset of 10 examples:

Example	Type of EV	Charging Infrastructure	Range	Purchased EV?
1	Compact	Home Charging	Short Range	Yes
2	Sedan	Public Charging	Medium Range	No
3	SUV	Workplace Charging	Long Range	Yes
4	Compact	Public Charging	Medium Range	Yes
5	Sedan	Home Charging	Long Range	Yes
6	SUV	Public Charging	Short Range	No
7	Compact	Workplace Charging	Long Range	Yes
8	Sedan	Home Charging	Short Range	No
9	SUV	Public Charging	Medium Range	Yes
10	Compact	Workplace Charging	Medium Range	No

Calculate the probability of a customer purchasing an electric vehicle given that:

1. The EV is a Sedan, using Public Charging, and has a Medium Range.
2. The EV is an SUV, using Home Charging, and has a Short Range.
3. The EV is a Compact, using Workplace Charging, and has a Long Range.

Provide your answers rounded to two decimal places.

Rajiv Gandhi Institute of Petroleum Technology

(An Institute of National Importance)

Mid-Semester Examination, Odd Semester, 2025-2026

B. Tech. 4th YEAR (CSE + IT + CSD + IDD)

Roll No. 2242092

Course Name : Generative Artificial Intelligence
Course Code : CS441

Time: 2 Hours
Max. Marks: 30

Instructions:

1. Do not write anything other than your roll number on the question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Q1: Explain the core idea of adversarial training in Generative Adversarial Networks with its [10] architecture and training objectives. Justify why GAN training is expressed as a min-max optimization problem. Suppose in a GAN training, a mini-batch for an iteration is given as:

- Real samples: $D(x) = [0.9, 0.8, 0.7, 0.95]$
- Generated samples: $D(G(z)) = [0.2, 0.3, 0.1, 0.4]$

Considering the above, solve the following questions:

1. Compute the discriminator loss L_D for this batch
2. Compute the generator loss L_G for the output sample.
3. Interpret what these values imply about the discriminator and generator's performance at this training step.

OR

Q1. Explain the major challenges of non-convergence and mode-collapse faced during the training [10] of Generative Adversarial Networks, illustrating with numerical examples. Discuss in detail any two GAN variants, proposed to overcome the mentioned challenges.

Q2: Why are VAEs considered generative models, while vanilla autoencoders are not? What is [10] the reparameterization trick in VAE, how and why is it implemented? Given $\mu = [0.5, -1.0]$, $\sigma = [0.2, 0.5]$ and a random sample $\epsilon = [0.1, -0.3]$ from $N(0,1)$, compute the latent vector z using reparameterization trick. Assume the parameters whenever required.

OR

Q2. Why does VAE use $KL(q(z|x) || p(z))$ instead of $KL(p(z) || q(z|x))$? The figure below [10] illustrates the idea of **variational inference (VI)**. The ellipse represents a family of approximate distributions $q(z;v)$ parameterized by v . The black dot outside the ellipse corresponds to the true posterior $p(z|x)$. Optimization starts from v_{init} and converges to v^* , the optimal parameters within the family. Illustrate this process of the VI and answer the questions below:

they recorded a short video message so that communication was not delayed and no one was excluded.

Another key learning was the importance of staying visible when working remotely. Team members made an effort to share their ideas during virtual meetings. HubSpot also used Slack to communicate across teams. By participating in Slack discussions, whether about work tasks or casual topics, employees remained connected and visible to colleagues working in different locations.

According to Rachel, Remote Week was a success. The team improved collaboration across locations and felt more confident that everyone on the team had an opportunity to be heard.

Questions:

(a) Using Solomon Asch's experiment, analyze why some team members may hesitate to speak up in virtual meetings. How did HubSpot's "Remote Week" practices help address this issue?

(10)

(b) Which leadership style is most evident in the HubSpot "Remote Week" initiative? Justify your answer with examples from the case. (10)

(c) How can the Johari Window be used to improve openness and trust in virtual teams?

(10)



22.09.2022

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
(An Institution of National Importance established under the Act of Parliament)

Mid Semester Examination

Seventh Semester (Odd Semester) – B. Tech – CSE+IT+M&C+EEE+EC&EV+CSD & IDD

Ethics and Corporate Governance

Max. Marks: 30

Duration: 120 Minutes

Date: 23.09.2025

Time: 12.00 PM – 02.00 PM

ANSWER ALL THE QUESTIONS

Section A ($3 \times 2 = 6$ marks)

1	Write any two principles of Business Ethics.
2	Differentiate between Personal Ethics and Professional Ethics
3	Write a short note on Whistle-blowing.

Section B ($4 \times 6 = 24$ marks)

4	What is CSR? Explain Carroll's CSR Pyramid and its importance.
5	How should managers handle ethical dilemmas? Explain with an example.
6	Discuss the Ethical Theories of Utilitarianism, Justice, and Rights.
7	Explain Stakeholder Mapping and Triple Bottom Line with examples.

Rajiv Gandhi Institute of Petroleum Technology

Roll No. 21CS2002

(An Institute of National Importance)

End-Semester Examination, Odd Semester, 2025-2026

B. Tech. 4th YEAR (CSE + IT + CSD + IDD + MNC + Minor)

Course Name : Computer Vision and Pattern Recognition
Course Code : CS454

Time: 3 Hours
Max. Marks: 60

Instructions:

- Do not write anything other than your roll number on the question paper.
- Assume suitable data wherever essential and mention it clearly.
- Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

SECTION: A

- Q1. (a) (5 marks) Explain in detail the Horn-Schunck optical flow algorithm, starting from the optical flow constraint equation $I_x u + I_y v + I_t$ to the iterative update equations for u and v .
- (b) (2 marks) Discuss the effect of the smoothness parameter λ , on the updated values
- (c) (5 marks) Compute u_{new} and v_{new} (showing all steps) using the Horn-Schunck update rule on the given values below:

$$I_x = 1, \quad I_y = 2, \quad I_t = -3$$

$$\bar{u} = 0.2, \quad \bar{v} = -0.4, \quad \lambda = 1$$

- (d) (3 marks) Explain the difference between the true motion field and optical flow. Why does optical flow not always represent real motion?

- Q2. (a) (5 Marks) Explain how the Kalman filter is used to track a moving feature point in an image sequence. In your explanation, clearly describe:

- State model, s_t
- measurement model, z_t
- roles of covariance, Σ_t
- Kalman gain, K_t
- The prediction-update cycle.

	c) Marketing strategy formulation d) Lobbying with regulators	
10.	An ESG-focused investor is most likely to reject an investment in a company that: a) Regularly publishes integrated reports and climate-risk disclosures b) Has strong boards but faces repeated penalties for labour-rights violations c) Has moderate profits but a clear decarbonisation roadmap d) Engages with stakeholders on diversity and inclusion	1
SECTION B		
Attempt ALL Parts and Answer in short.		(5x4)
1.	Define business ethics. Discuss the difference between personal ethics and professional ethics with suitable examples.	4
2.	Briefly explain any two ethical theories (e.g., Utilitarianism, Rights, Justice) and illustrate how they influence business decision making.	4
3.	Write a short note on Carroll's Pyramid of CSR and explain how it connects with the Triple Bottom Line (People, Planet, Profit).	4
4.	Explain any four determinants of corporate governance (internal or external) and their impact on the quality of governance in a company.	4
5.	What are the key provisions of CSR under Section 135 of the Companies Act, 2013? Discuss eligibility criteria and CSR committee requirements in brief.	4
SECTION C		
Attempt ALL Parts		(2x10)
1.	a) Critically discuss how AI can support corporate governance and ESG reporting and what ethical concerns arise due to the use of AI. Suggest principles for ethical use of AI in governance. b) What is an ethical dilemma in business? Describe a realistic ethical dilemma that could occur in Business, HR, Marketing, and Finance.	5+5
2.	A large listed company has been accused of manipulating its financial statements to show inflated profits over several years. Later, it is discovered that senior management had suppressed information from the Board and auditors. a) Identify and discuss the ethical issues and governance failures in this scenario. b) Who are the key stakeholders affected and how? c) Suggest a set of corporate governance and internal control measures that could help prevent such scandals.	5+2+3

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B. Tech./IDD Mid Semester Examination (Odd Sem, 2025-26)

Subject: Principles of Economics

Time: 2hrs

Full Marks: 30

Answer all the questions. Use diagrams wherever possible to support and explain your answers clearly

- Q1.** Divya quits her job, which pays \$30,000 a year, to finish her college degree. Her annual college expenses are \$10,000 for tuition, \$2,000 for books, and \$700 for food. What is her opportunity cost of attending college for the year?
- Q2.** A German worker takes 400 hours to produce a car and 2 hours to produce a case of wine. A French worker takes 600 hours to produce a car and X hours to produce a case of wine.
- a. For what values of X will gains from trade be possible? Explain.
- b. For what values of X will Germany export cars and import wine? Explain.
- Q3.** Are the following statements true or false? Explain in each case.
- a) "Two countries can achieve gains from trade even if one of the countries has an absolute advantage in the production of all goods."
- b) "Certain very talented people have a comparative advantage in everything they do."
- c) "If a certain trade is good for one person, it can't be good for the other one."
- d) "If a certain trade is good for one person, it is always good for the other one."
- e) "If trade is good for a country, it must be good for everyone in the country."
- Q4.** Beer and pizza are complements because they are often enjoyed together. When the price of beer rises, what happens to the supply, demand, quantity supplied, quantity demanded, and price in the market for pizza?
- Q5.** Over the past 20 years, the cost of solar panels has fallen sharply due to advances in production technology. How do you think this has affected the market for electricity from solar power? For coal-based electricity? For solar installation services?
- Q6.** Consider the markets for *streaming subscriptions (like Netflix), smart TVs, and movie theatre tickets*.
- a. For each pair, identify whether they are *complements or substitutes*:
- Streaming subscriptions and smart TVs
 - Streaming subscriptions and movie tickets
 - Smart TVs and movie tickets
- b. Suppose a technological advance reduces the cost of manufacturing smart TVs. Draw a diagram to show what happens in the market for smart TVs.
- c. Draw two more diagrams to show how the change in the market for smart TVs affects the markets for streaming subscriptions and movie tickets.
- Q7.** Consider the following events: Scientists reveal that eating oranges decreases the risk of diabetes, and at the same time, farmers use a new fertilizer that makes orange trees produce more oranges. Illustrate and explain what effect these changes have on the equilibrium price and quantity of oranges.
- Q8.** For each of the following pairs of goods, which good would you expect to have more elastic demand and why?
- a. required textbooks or mystery novels
- b. iPhones vs. smartphones in general
- c. Netflix subscription vs. streaming services in general
- d. Coffee today vs. coffee over the next two years
- Q9.** The weekly demand schedule for *online movie rentals* in a city is given below:
- | Price (₹) | 60 | 50 | 40 | 30 |
|----------------------------------|----|----|----|----|
| Quantity Demanded ('000 rentals) | 20 | 30 | 45 | 70 |
- a. Using the midpoint (arc) method, calculate the *price elasticity of demand* between:
- i) ₹60 and ₹50 ii) ₹50 and ₹40 iii) ₹40 and ₹30
- b. For each range, classify demand as *elastic, inelastic, or unit elastic*.
- c. Compute *total revenue* ($TR = P \times Q$) at each price level. Based on your elasticity results, explain how elasticity is related to changes in total revenue.
- d. Draw a well-labelled *demand curve diagram*, marking the three segments clearly.

SECTION: B

- ❖ Each question in SECTION B carries 1 mark, for a total of 30 marks across 30 MCQs.
- ❖ Answer by writing only the correct option (A, B, C, or D) for each question in order

Q1. Horn-Schunck optical flow algorithm:

$$u^{new} = \bar{u} - \frac{I_x(I_x\bar{u} + I_y\bar{v} + I_t)}{\alpha^2 + I_x^2 + I_y^2}$$

If $\alpha \rightarrow \infty$:

- A. data term dominates
- B. smoothness dominates
- C. optical flow disappears
- D. edges sharpen

Q2. In Lucas-Kanade:

$$u = (A^T A)^{-1} A^T b$$

- A. solution is unique
- B. system is stable
- C. aperture problem occurs
- D. noise causes instability

Q3. In optical flow:

$$I_x u + I_y v + I_t = 0$$

- A. Flow is zero
- B. Flow cannot be determined
- C. Flow is infinite
- D. Flow equals

Q4. At a pixel location, you are given: $I_x=2$, $I_y=1$, $I_t=-3$. If the horizontal motion $u=1$, compute the vertical component v using the Optical Flow Constraint Equation.

- A. 2
- B. -3
- C. -1
- D. 1

Q5. The Shi-Tomasi corner detection is used to:

- A. Detect moving edges
- B. Select strong feature points for tracking
- C. Reduce image size
- D. Smooth out the motion

Q6. The Horn-Schunck energy functional includes which two terms?

- A. Brightness constancy and small motion
- B. Data term and smoothness term
- C. Edge term and region term
- D. Gradient term and texture term

Exhibit 8: Conversion Assumptions

- 1 tonne = 1,000 kg
- All emissions must be reported in kg CO₂e per scooter
- Full load utilization is assumed for all transport modes
- Warehousing and inventory emissions are ignored

Answer the following questions:

- ✓ Draw a clearly labeled global supply chain map showing manufacturing locations and transport modes
- ✓ Calculate total manufacturing-related GHG emissions per scooter using Exhibits 2–4. Show all steps.
- ✓ Calculate transportation emissions for all inbound and outbound logistics using Exhibits 5–7.
- ✓ Compute the total cradle-to-gate carbon footprint per scooter.
- ✓ Identify the single largest emission contributor and suggest two supply-chain-level mitigation strategies.

Mini case-2. Bharat Electronics Limited (BEL) is a leading Indian public-sector enterprise engaged in the design, manufacture, and lifecycle support of defence electronics such as radar systems, communication equipment, avionics modules, electronic warfare systems, and surveillance electronics. 10

BEL's typical value chain includes:

- *Inbound logistics:* sourcing semiconductors, printed circuit boards (PCBs), rare earth magnets, specialty metals, and chemicals from domestic and global suppliers.
- *R&D and design:* high-precision electronic design, prototyping, and software integration.
- *Manufacturing & testing:* PCB assembly, soldering, coating, machining of enclosures, clean-room testing, and quality assurance—energy-intensive processes with use of solvents, chemicals, and water.
- *Outbound logistics:* transportation of sensitive equipment to defence establishments across India, often involving secure and specialized logistics.
- *Installation, maintenance & end-of-life:* long product life cycles (15–25 years), upgrades, refurbishment, and eventual disposal of obsolete electronic systems.

Key sustainability challenges faced by BEL include high electricity consumption, generation of electronic and hazardous waste, use of conflict-sensitive minerals, worker health and safety in high-precision manufacturing, and increasing regulatory scrutiny on ESG disclosures.

BEL's Board has decided to adopt the SDG Compass framework to integrate sustainability into its core business strategy. Answer the following, following the **SDG Compass** steps:

- ✓ a) Based on BEL's operations and impact areas described above, identify four UN Sustainable Development Goals (SDGs) that are most relevant to BEL. Explain and justify!
- ✓ b) For the four SDGs identified in part (a), map each SDG to the most appropriate value-chain stage(s) of BEL (inbound logistics, R&D, manufacturing, outbound logistics, or after-sales/end-of-life).
Clearly indicate where BEL's activities create the strongest positive or negative impact for each SDG.
- ✓ c) Using any two prioritization criteria recommended under the SDG Compass (such as severity of impact, stakeholder expectation, regulatory risk, or strategic relevance), rank the four SDGs from highest to lowest priority for BEL.
Provide a brief justification for your prioritization.
- ✓ d) For the top two priority SDGs, propose one concrete, actionable initiative per SDG that BEL could implement. *Each initiative should specify: the value-chain stage targeted; the specific sustainability issue addressed; and how the initiative also strengthens BEL's long-term competitiveness or compliance posture.*

- (b) Do you think employees have a responsibility to be careful about what they blog about their company? (5)
 - (c) Do you think that the employees who blog about their companies have an ethical responsibility to disclose their identities? Why or why not? (5)
 - (d) How can a company develop a policy for handling communication of sensitive issues inside, and outside, the company? (5)
2. Read the caselet and answer the questions that follow:

The Manager's Memo

Following is the letter authored by the Vice President, Consumer Electronics Division and addressed to the Production Manager. The issue raised in the letter relates to Quality First Programme.

"I think it's time to look at a new approach to our Quality First Programme. We've had the suggestion box out for a year and so far we've received only two suggestions for quality improvements, of which only one made any sense."

"Let's try a group approach. We can divide the employees into groups that can meet to come up with ideas for improving the quality of our products and production process."

"I'd like you to prepare a plan for these groups. How should they operate? How often should they meet? What should be the size of each group? Should we divide the groups into engineers, production workers, maintenance personnel and so on, or should we put a variety of workers into each group? What problems can we anticipate with the group format, and how can we resolve them? These questions are meant as a starting point. If you have other ideas or concerns, please include them in your plan."

Questions

- (a) Assume you are the production manager and respond to the Vice President's letter with a memo outlining the plan requested. Be sure to respond to the specific questions. Use the concepts discussed in class to develop a complete plan that minimizes the pitfalls of group dynamics. (10)
3. Among the major leadership theories, which one, according to you, offers the strongest explanatory and practical value in contemporary organizational settings? Critically justify your answer. (10)
4. Critically evaluate the Johari Window as a framework for understanding self-awareness and interpersonal dynamics in an organization. (10)

	(b) Ignore the issue and do all the work yourself (c) Have a private discussion to understand reasons, clarify expectations, and document next steps (d) Shout at the member in front of the whole class	
5.	During a virtual Group Discussion on "Future of Remote Work," what behavior will most strongly reflect leadership and industry readiness? (a) Keeping your camera off and speaking only in the last minute (b) Dominating the conversation and not allowing others to speak (c) Bringing 2-3 data-backed points, building on others' inputs, and helping the group reach a conclusion (d) Criticizing the moderator's topic selection	1
6.	Your manager informs you that your current role will soon be impacted by automation using AI tools. Which response best demonstrates employability in an Industry 4.0 context? (a) "This is unfair; I will only do the same tasks as before." (b) "Can you guide me on which tools and skills I should start learning so that I can contribute to the new workflows?" (c) "This is not my responsibility; let others adapt." (d) "I will work only if there is no AI tool involved."	1
7.	In your mid-year review, your supervisor writes: "Strong technical understanding, but needs improvement in stakeholder communication and documentation." What is the best way to convert this into an action plan? (a) Ignore it because ratings are already submitted (b) Ask for 1-2 concrete examples and request suggestions for courses/practices to improve communication and documentation (c) Argue that communication is not important for engineers (d) Blame your team members for not supporting you	1
8.	A company uses Applicant Tracking Systems (ATS) to shortlist CVs. Which CV version is most likely to pass ATS screening for a "Junior Data Analyst" role? (a) CV with heavy graphics, text inside images, and no clear headings (b) Plain text CV with clear sections, bullet points mentioning "Excel, SQL, basic Python, dashboard creation," and project details (c) CV written entirely in paragraph form without any keywords from the job description (d) One-page CV with only your personal details and hobbies	1
9.	During a critical client demo, your code suddenly fails 10 minutes before the meeting. What is the most employable reaction? (a) Hide the issue and hope no one notices (b) Inform your senior/manager immediately, suggest a backup plan (recorded demo or stable older version), and start root-cause analysis after the meeting	1

8.	Least privilege in access control means: a) Everyone is admin b) Grant only minimum required access c) Access is permanent d) No audits	1
9.	The strongest impact statement is: a) "Used XYZ tool extensively" b) "Learned a lot" c) "Reduced cycle time by 22% by automating the test/data pipeline" d) "Faced many challenges"	1
10.	An effective elevator pitch should be: a) 30–60 s: role, strengths, value proposition b) 5–7 min c) A course list d) A joke	1

SECTION B

Attempt **ALL** Parts and Answer in short (About 90 to 120 words). **(4x3)**

1.	A graduate role includes problem-solving, working with others, talking to people who aren't technical, being able to read and understand data, using basic tools (like programming, CAD/EDA, lab instrumentation, or analytics), and keeping track of versions and changes. - Write down five words or phrases that you would naturally use in your résumé or profile. - Add one measurable success that fits with these keywords.	3
2.	Write a short STAR note about how you reduced a delivery risk (the risk of missing a deadline) in your field (software feature, PCB spin, test rig, data model, or literature review). 1–2 lines for each S, T, A, R, with a number at the end to show how well it worked.	3
3.	You have three things to do today: (A) fix a P1 problem that is affecting users and tests, (B) get ready for tomorrow's review slides, and (C) On-board a new team member/intern. - Give a list of priorities with reasons (impact, urgency, dependencies). - Send one short message to all stakeholders letting them know your plan and any requests or escalations.	3
4.	Rewrite this request for a 24-hour extension into a professional email with a good reason and a new plan: "Hey, I can't finish the report because I'm so busy." Will send later.	3

SECTION C

Write your answer by choosing your context with any one of the following:

- Software/System Feature** (service/API, firmware module, UI, or control algorithm)



Rajiv Gandhi Institute of Petroleum Technology
Bengaluru Campus
End-Semester Examination
(ODD Semester, 2025-26)

Max. Marks: 50 **Duration:** 03 Hrs
Programme/Semester: B. Tech 7th Semester
Course Title: Ethics and Corporate Governance

Date: 05/12/2025 **Roll No:** 22 4 2002
Time: 11:30 PM-02:30 PM

S.NO	Questions	Marks
SECTION A (Objectives)		
Attempt ALL Parts and select the best option		(10x1)
1.	A listed company is offered a highly profitable project that clearly violates environmental emission norms, but non-compliance is unlikely to be detected in the short run. A deontological (duty-based) approach to ethics would most likely recommend: a) Accepting the project because it maximises shareholder wealth b) Accepting the project but creating a CSR programme to offset the damage c) Rejecting the project because violating a rule is wrong irrespective of outcomes d) Postponing the decision until public pressure increases	1
2.	Under stakeholder theory, which of the following decisions is most consistent with the manager's obligations? a) Closing a plant without notice to reduce costs and boost quarterly EPS b) Sharing advance lay-off plans with employees and local authorities and creating a transition package c) Ignoring community protests as long as the decision is legal d) Prioritising only the interests of institutional investors	1
3.	A company introduces an "ethics hotline" operated by a third-party firm, allows anonymous reports, and protects employees from retaliation when they report misconduct. These mechanisms are primarily meant to strengthen: a) Aggressive marketing strategy b) Whistle-blower protection and ethical culture c) Price leadership in the market d) Tax planning efficiency	1
4.	Which one of the following situations best illustrates greenwashing? a) A company genuinely achieving carbon neutrality and publishing audited reports b) A company paying all environmental fines on time	1

Q7. Which of the following is an advanced large displacement optical flow method?

- A. Horn-Schunck
- B. MDP-Flow2
- C. Lucas-Kanade
- D. Shi-Tomasi

Q8. The term "DeepFlow" refers to:

- A. CNN-based deep optical flow
- B. Dense hierarchical matching with variational refinement
- C. Gradient-based estimation
- D. Edge flow analysis

Q9. The motion-aware piecewise parametric flow model belongs to:

- A. MDP-Flow2
- B. DeepFlow
- C. Lucas-Kanade
- D. Horn-Schunck

Q10. The covariance matrix in Kalman filtering represents:

- A. The uncertainty in state estimation
- B. The rotation of the camera
- C. The intensity gradient
- D. The sampling rate

Q11. The main limitation of the Kalman Filter in object tracking is:

- A. High training cost
- B. Failure under nonlinear or random motion
- C. Inability to model object appearance
- D. Requirement of large datasets

Q12. If process noise Q is increased (all else equal), the Kalman gain K in the next step will generally:

- A. Decrease (less weight to measurements)
- B. Increase (more weight to measurements)
- C. Become zero
- D. Remain unchanged

Q13. Suppose the measurement noise R becomes very large (very noisy measurements). The Kalman gain K tends to:

- A. A vector of zeros (trust prediction only)
- B. Infinity (unstable)
- C. Unchanged
- D. Equal to $[1, 0]^T$

Q14. Lucas-Kanade optical flow fails when:

- A. gradients are strong
- B. motion is constant
- C. texture is low or absent
- D. λ is large

Q15. Which method uses feature descriptors to match distant corresponding pixels?

- A. SIFT Flow
- B. Sobel Matching
- C. Derivative Matching
- D. Gaussian Tracking

- Q16. In Kalman filter, the state vector contains:
- A) Only position
 - B) Position + velocity
 - C) Image gradient values
 - D) Depth information at pixel

Q17. The factorization method for 3D reconstruction assumes:

- A) Large inter-frame motion
- B) Orthographic projection
- C) Known camera intrinsics
- D) Full-rank disparity

Q18. If l is the object size in pixels and $\dot{l}$ is its rate of change in pixels/second, Which statement of the following is TRUE Given

At time t_1 :

- $l = 60$ px
- $\dot{l} = 5$ px/s

Later at time t_2 :

- $l = 72$ px
- $\dot{l} = 6$ px/s

- A) TTC decreases from 12s to 10s
- B) TTC increases from 10s to 12s
- C) TTC stays the same
- D) TTC becomes infinite

Q19. DeepSORT improves upon SORT by integrating:

- A) Region proposal network
- B) HOG descriptors
- C) Convolutional appearance features
- D) Manual feature selection

20. An object's image width changes from 40 px to 44 px over 0.5 seconds. Assume linear growth in size. Approximate $\dot{l}$ and compute τ at the start.

- A) 4 px/s $\rightarrow \tau = 10$ s
- B) 8 px/s $\rightarrow \tau = 5$ s
- C) 4 px/s $\rightarrow \tau = 5$ s
- D) 8 px/s $\rightarrow \tau = 10$ s

21. The term $p(x | \omega_j)$ in Bayes' theorem is called the:

- A. Posterior probability
- B. Prior probability
- C. Likelihood function
- D. Conditional risk

22. If $P(\omega_1)=0.6$ and $P(\omega_2)=0.4$, and no feature information is available, what is the minimum probability of error?

- A. 0.4
- B. 0.5
- C. 0.6
- D. 0.2

23. The likelihood ratio test decides ω_1 if:

1. Read the case study and answer the question that follows:

Should Companies That Fire Shoot First

In the recessions in the early eighties and nineties and after the 2001 terrorist attacks, layoffs were fairly private affairs. To be sure, news would often leak out to local and national media outlets, but companies did their best to keep it as quiet as possible. One consequence of the growth of the Internet in general, and of social networking sites in particular, is that this is no longer possible.

When Microsoft's Hyderabad office laid off 1 percent of the workforce, employees were very upset and a few websites came into being. However, some companies are taking a more proactive approach. When Tesla Motors laid off its employees, its CEO, Elon Musk, posted a blog entry on the topic just before announcing the layoffs to employees. "We had to say something to prevent articles being written that were not accurate," he said.

"Today, whatever you say inside a company will end up in a blog," says Rusy Rueff, a former executive at PepsiCo. "So, you have a choice as a company—you can either be proactive and say, 'Here's what's going on,' or you can allow someone else to write the story for you."

Illustrating the perils of ignoring the blogosphere, when newspaper giant Gannett announced it was laying off 10 percent of its employees, it posted no blog entries and made no statement. Jim Hopkins, a 20-year veteran who left the company just before the layoffs, writes the unofficial Gannett Blog. "I try to give the unvarnished truth. I don't think the company offers the same level of openness to employees," he said. Gannett spokeswoman Tara Connell replied, "We attempt to make those personal communications happen as quickly as possible."

Says blog expert Andy Sernovitz, "There are rigid companies that still wish there was traditional P.R. control of the message, but the day is long over."

Questions

- (a) Evaluate blogging as a communication channel in organizations. What are its advantages and risks compared to formal communication channels? (5)



राजीव गांधी पेट्रोब्रियम प्रौद्योगिकी संस्थान, बंगलुरु परिसर
Rajiv Gandhi Institute of Petroleum Technology, Bengaluru Campus

(राष्ट्रीय महत्व का संस्थान) Institution of National Importance)

कंबलपुर, सुलीबेल हवेली, होसकोटे, बंगलुरु, कर्नाटक - 562165, भारत
Kambalipura, Sulibele Hobli, Hoskote, Bengaluru, Karnataka - 562165, India

Roll no. 2242002

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B.Tech IDD-22 Fourth Year Even Semester
Mid-Semester Examination 2025-26
Subject: MT5213 Sustainability and Climate Change

Time: 2 hrs.

Full Marks: 30

Note: Attempt all the questions. All questions carry equal marks
The use of a scientific calculator is allowed

1. A Assess the socio-economic consequences of global warming beyond environmental impacts. 3M
B Explain the carbon cycle with a neat diagram. 4M
C Evaluate whether global warming is solely a result of anthropogenic activities. Provide justification. 3M
2. A Explain the major international climate protocols and agreements aimed at addressing climate change. 4M
B Differentiate between the natural greenhouse effect and the enhanced greenhouse effect. Critically analyze why the latter is a concern. 3M
C Analyze how global warming can trigger feedback mechanisms in the climate system. 3M
3. Case Scenario 10M
By 2035, the average global temperature is projected to exceed 1.5°C above pre-industrial levels, according to assessments by the Intergovernmental Panel on Climate Change. A middle-income country with a 40% agriculture-dependent population is facing repeated droughts, declining groundwater levels, and frequent cyclones affecting coastal cities.
Agricultural productivity has declined by 20% over a decade, increasing food prices and rural unemployment. Urban flooding has damaged infrastructure, causing economic losses equivalent to 3% of GDP annually. Health authorities report a rise in vector-borne diseases and heat-related mortality.

Q3. In Region-based Convolutional Neural Networks (R-CNN), explain how the CNN deals with [10] variable input sizes and converts them into fixed-size feature vectors. Consider a Faster R-CNN model that processes an input image of size 32×32 pixels using a convolutional feature extractor with: Kernel size = 5, Stride = 5, No padding. The model generates 9 anchor boxes per feature map location, using 3 different scales and 3 aspect ratios. The Region Proposal Network (RPN) evaluates each anchor by:

- Assigning a positive label if $\text{IoU} \geq 0.7$ with a ground-truth box
- Assigning a negative label if $\text{IoU} \leq 0.3$
- Discarding anchors with intermediate IoU ($0.3 < \text{IoU} < 0.7$)

Out of all generated anchors:

- 324 anchors are produced in total
- 30 anchors are labeled positive, 70 negative, and the rest are ignored
- After scoring and applying Non-Maximum Suppression (NMS), 128 proposals are selected for ROI classification

Considering the above, perform the following calculations (Assume parameters if required):

- Calculate the size of the output feature map after convolution.
- Calculate the total number of anchors generated.
- What percentage of anchors are labeled: (i) positive, (ii) negative, (iii) ignored?
- If the RPN takes 40 ms to score all anchors and generate proposals, what is the average time per anchor in microseconds (μs)?
- Calculate the percentage of anchors retained after NMS.

OR

Q3. Explain how YOLO divides the input image into a grid and assigns responsibilities to grid [10] cells and anchor boxes for detecting objects. What problems may arise from this grid-based assignment. Consider an image of size 416×416 pixels is input to a YOLOv3 model. The model divides the image into a 13×13 grid and uses 3 anchor boxes per grid cell. The following ground truth bounding boxes are present in the image:

Object	Center (x, y) in pixels	Width (w)	Height (h)
A	(105, 220)	80	100
B	(200, 80)	150	160
C	(410, 410)	30	40

The anchor box dimensions (in pixels) are:

Anchor	Width	Height
A1	80	100
A2	100	150
A3	50	60

Based on above, perform the following tasks:

- For each object (A, B, C), determine the grid cell (i, j) that will be responsible for detecting the object.
- Determine which anchor box (A1, A2, or A3) will be assigned to each object using IoU (assume all anchors and boxes are centered in the same grid cell).

The firms make decisions *simultaneously*, and their profits depend on the combination of strategies chosen. The estimated profit payoffs (in ₹ crores) are shown below:

Payoff Matrix

	MobiQuest: H	MobiQuest: L
TechNova: H	(70, 70)	(40, 85)
TechNova: L	(85, 40)	(55, 55)

- Identify each firm's *best response* to the rival's pricing strategy. Show your reasoning using the payoff table.
- Determine the **Nash equilibrium** (if any). Clearly explain why this outcome is stable.
- Does either firm have a **dominant strategy**? Explain.
- Suppose the two firms can form a collusive agreement to jointly maximise industry profits.
 - What pricing combination would they prefer under collusion?
 - Compare this with the Nash equilibrium and briefly explain why the market outcome differs from the collusive outcome.

Q6. Nimbus Coffee Roasters Pvt. Ltd. is a mid-scale specialty coffee producer supplying premium roasted coffee beans to cafés across Bengaluru and Hyderabad. The firm operates a small roasting facility where labour and utility costs vary with output, while the monthly rental and machine depreciation remain fixed. Management wants to evaluate how costs behave as the firm scales production. You are hired as a junior business analyst to compute key cost measures and advise on efficiency and pricing decisions.

Dataset (per week)

The table below shows weekly output levels and **Total Cost (TC)** recorded by the operations team:

Output (kg)	Total Cost (₹)
0	12,000
10	18,000
20	24,000
30	29,000
40	35,000
50	43,000

- Calculate the following at each positive output level (10 to 50 kg):

- Average Fixed Cost (AFC)
- Average Variable Cost (AVC)
- Average Total Cost (ATC)
- Marginal Cost (MC)** between each output increment.

Use standard cost formulas and show all steps.

- Based on your results, answer the following in brief:

- At what output level does Nimbus Coffee achieve the **lowest ATC**?
- Explain why **MC intersects AVC and ATC at their minimum points**, using your calculated numbers as evidence.
- If the market price per kg of roasted coffee is ₹900, should the firm produce 40 kg or 50 kg? Justify using MC and ATC.

Q7. A city is evaluating the impact of congestion caused by e-scooter deliveries. The **demand curve** for delivery trips is: $P = 40 - 0.1Q$

Private marginal cost (PC) is constant at ₹10 per trip.

Each trip creates an external cost of ₹5, so the **social cost (SC)** is ₹15.

Draw a clearly labelled diagram showing Demand, PC, SC, Q_{market} , and Q_{social} .

- Demand or marginal private benefit (MPB)
- Marginal Private Cost (MPC) (horizontal at ₹10)
- Marginal Social Cost (MSC), drawn above MPC (horizontal at ₹15)
- Q_{market} and Q_{social}

- Compute the market equilibrium quantity of trips.
- Compute the socially optimal quantity of trips.
- State the Pigovian tax that would achieve the social optimum.
- Calculate the deadweight loss (DWL) created by overproduction in the unregulated market.

$$DWL = \frac{1}{2} \times (Q_{\text{market}} - Q_{\text{social}}) \times \text{External cost}$$

Q4. (5 Marks) Consider the following two 8×8 images. Perform histogram matching for the image on the left using the reference image on the right and plot the histogram of the reference image and the resulting histogram of the matching.

0	5	7	7	5	8	7	8
7	2	6	2	6	5	6	8
6	9	7	7	0	7	2	7
6	6	1	7	6	7	7	5
9	6	0	7	8	2	6	7
2	8	8	2	7	6	7	8
7	3	2	6	1	7	5	8
9	9	5	6	7	7	7	7

1: Original Image

2	6	7	8	5	0	6	8
7	2	6	2	6	2	6	7
6	8	9	2	0	7	3	7
7	6	1	4	6	0	5	9
9	7	0	9	8	4	6	7
9	8	9	4	7	1	7	8
7	3	5	5	1	3	5	7
5	8	8	4	6	2	7	6

2: Reference Image

Q5. (3 Marks) An image has gray levels $r \in [0,1]$ with probability density function:

$$p_r(r) = e^{-r}, \quad 0 \leq r \leq 1$$

We want the output image to have the specified histogram:

$$p_z(z) = 2z, \quad 0 \leq z \leq 1$$

Find the gray-level transformation function $z = G^{-1}(T(r))$

Q6. (2 Marks) Given the following 5×5 grayscale neighborhood centered at pixel $f(x,y)$:

12	18	20	22	15
14	16	19	21	13
17	200	25	23	18
11	15	20	24	16
10	14	18	19	12

Find the new value of the center pixel using the following filters:

- Median filter
- Alpha-trimmed mean filter with $d=2$

Q7. (5 Marks) The following 5×5 neighborhood represents a small portion of a grayscale image:

48	50	49	52	51
47	48	50	51	52
49	50	140	52	53
50	51	52	53	54
51	52	53	55	56

Calculate the values of the image after applying:

- Sobel Filter
- Laplacian Filter
- Unsharp Masking
- High-Boost Filtering ($k = 2$)
- Compare the observations for detecting an edge. (Assume the parameters wherever needed)

Q8. (5 Marks) Justify and derive the gray-level transformation functions of histogram equalization and histogram specification as contrast enhancement methods.

Rajiv Gandhi Institute of Petroleum Technology

Roll No. 22182002

(An Institute of National Importance)

End-Semester Examination, Odd Semester, 2025-2026

B. Tech. 4th YEAR (CSE + IT + CSD + IDD)

Course Name : Generative Artificial Intelligence
Course Code : CS441

Time: 3 Hours
Max. Marks: 50

Instructions:

1. Do not write anything other than your roll number on the question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

SECTION A (25 Marks)

1. Consider a transformer model with standard sinusoidal positional encoding. Answer the following (Assume the variables wherever necessary) :
 - a. (2 Marks) Discuss the properties that an ideal positional encoding scheme should satisfy.
 - b. (2 Marks) For a transformer with embedding dimension $d=512$, justify and derive the expression for the k -th frequency ω_k .
 - c. (2 Marks) Explain how the frequency ω_k determines the relative wavelength of that dimension and how this influences positional representation.
 - d. For an embedding dimension $d=8$, perform the following computations considering the sinusoidal positional encoding:
 - i. (2 Marks) Compute all the frequencies ω_k .
 - ii. (2 Marks) Write the full positional encoding vector P_5 .
 - iii. (2 Marks) Comment on how the different frequencies affect sensitivity to local vs. global positional changes in the sequence.
2. Considering a transformer architecture with multi-head scaled dot-product attention. Answer the following (Assume the variables wherever necessary) :
 - a. (2 Marks) Explain why the Transformer uses multiple attention heads instead of a single attention mechanism.
 - b. (3 Marks) Describe step by step with formulation on how a single attention head computes attention.

Rajiv Gandhi Institute of Petroleum Technology, Bangalore

B. Tech. Winter Term

Mid-Semester Examination 2025

Subject: HU331 Organizational Psychology

Note: Attempt all questions

Time 2 hrs

Full Marks: 30

- ✓ 1. A luxury resort introduces a new "Guest Experience Excellence Program" that requires all staff to follow detailed emotional display rules—maintaining constant warmth, avoiding visible frustration, and responding empathetically even to abusive guests. Employees attend training sessions that focus on emotion regulation techniques but offer little discussion on emotional boundaries.

Staff members report that while guest satisfaction scores have improved, emotional exhaustion and turnover have also increased.

Discussion Questions:

- ✓ (a) What is meant by Emotional Intelligence? (4)
- ✓ (b) To what extent is emotional intelligence distinct from impression management and role enactment in organizational settings? Can scripted empathy still be considered Emotional Intelligence? Why or why not? (6)
- ✓ 2. While there are similarities between Maslow's need hierarchy and Herzberg's two-factor theory, there is also an important difference. Maslow believes that the first three needs as motivators, whereas Herzberg believes they are hygiene factors. Which view do you agree with and why? (10)
- ✓ 3. Fill in the blanks: (5x2=10)
- ✓ (a) Locke's Goal-Setting Theory states that performance improves when goals are clear / specific, challenging, and accepted.
- ✓ (b) An individual who believes outcomes are controlled by luck has an external locus of control.
- ✓ (c) Information known to others but not to oneself lies in the blind spot Area of Johari Window.
- ✓ (d) A person who refuses immediate pleasure because it violates moral standards is experiencing conflict between the id and superego.

WCM / VERY LESS ?

Rajiv Gandhi Institute of Petroleum Technology, Bangalore

B. Tech. Winter Term

Mid-Semester Examination 2025

Subject: HU331 Organizational Psychology

Note: Attempt all questions

Time 2 hrs

Full Marks: 30

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- (c) Information known to others but not to oneself lies in the blind Area of Johari Window.
- (d) A person who refuses immediate pleasure because it violates moral standards is experiencing conflict between the super and ego.

(b) (10 Marks) Consider a 1D constant-velocity Kalman filter used to track the horizontal position of a feature point, with:

State at time t (a posteriori):

$$s_t = \begin{bmatrix} x_t \\ v_t \end{bmatrix} = \begin{bmatrix} 10 \\ 1 \end{bmatrix}$$

Covariance:

$$\Sigma_t = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

State transition:

$$\Phi = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Measurement matrix:

$$H = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Process noise:

$$Q = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

Measurement noise:

$$R = 1$$

Measurement at $t + 1$:

$$z_{t+1} = 11.2$$

Perform one full Kalman filter iteration to compute the following:

1. s_{t+1}^- (predicted state)
2. Σ_{t+1}^- (predicted covariance)
3. K_{t+1} (Kalman gain)
4. s_{t+1} (updated state)
5. Σ_{t+1} (updated covariance)

	c) A company heavily advertising a "green" product while its overall operations remain highly polluting d) A company investing in new clean-technology R&D	
5.	A firm designs its ethics programme mainly to avoid penalties and satisfy legal requirements (compliance checklists, mandatory training, fear of punishment). This is an example of a/an: a) Integrity-based ethics programme b) Compliance-based ethics programme c) Value-centric self-governance model d) Philanthropy-driven ethics system	1
6.	An Indian listed company has one-third independent directors, a separate Chairperson and CEO, an Audit Committee, and a Nomination & Remuneration Committee. These features primarily aim to: a) Reduce product development time b) Minimise agency problems and improve board oversight c) Increase the legal immunity of executives d) Replace external auditors	1
7.	A company subject to CSR provisions spends the required amount every year, but chooses projects only to maximise visibility and publicity, with minimal impact assessment or stakeholder consultation. Ethically, the major weakness of such an approach is that: a) It violates the Companies Act, 2013 automatically b) It focuses on image-building rather than genuine social responsibility c) It eliminates all financial benefits of CSR d) It is incompatible with any form of philanthropy	1
8.	In an AI-driven fraud detection system used by a bank, the model flags far more transactions from a particular community as "suspicious" despite similar behaviour across all customers. From an ethics and governance perspective, the primary concern is: a) Excessive transparency of the model b) Algorithmic bias and unfair discrimination c) Insufficient profit maximisation d) Over-compliance with regulations	1
9.	An Audit Committee insisting on rotation of audit partners, critical review of related-party transactions, and direct access to internal auditors is mainly strengthening which governance dimension? a) Board-employee informal relations b) Financial reporting integrity and internal control	1

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B. Tech./IDD Mid Semester Examination (Winter Sem, 2025-26)

Subject: **Sustainability & Climate Change**

Time: 2hrs

Full Marks: 30

Mini Case 1: Carbon Footprint Assessment of a Global Supply Chain

Aurora Mobility Pvt. Ltd. (AMPL) is an Indian firm assembling and selling a mid-range electric scooter for urban commuting. The company operates a globally distributed supply chain, sourcing major components from multiple countries and assembling the final product in India.

20

In response to ESG disclosure requirements and regulatory pressure, AMPL intends to estimate the *cradle-to-gate* carbon footprint (kg CO₂e per scooter). The analysis excludes use-phase and end-of-life emissions. Students are required to map the supply chain and compute emissions arising from manufacturing energy consumption and transportation.

Exhibit 1: Global Supply Chain Structure

Stage	Activity	Location
1	Battery cell manufacturing	Shenzhen, China
2	Electric motor manufacturing	Stuttgart, Germany
3	Aluminum frame manufacturing	Ho Chi Minh City, Vietnam
4	Final assembly and testing	Pune, India
5	Domestic distribution warehouse	Mumbai, India

Exhibit 2: Manufacturing Energy Consumption per Unit

Component	Electricity (kWh/unit)	Thermal Energy (MJ/unit)	Thermal Fuel
Battery pack	120	300	Natural Gas
Electric motor	85	180	Natural Gas
Aluminum frame	60	420	Fuel Oil
Final assembly	25	90	Natural Gas

Exhibit 3: Electricity Grid Emission Factors

Country	kg CO ₂ e / kWh
China	0.70
Germany	0.40
Vietnam	0.58
India	0.82

Exhibit 4: Thermal Fuel Emission Factors

Fuel Type	kg CO ₂ e / MJ
Natural Gas	0.056
Fuel Oil	0.077

Exhibit 5: Transportation Distance Matrix

From	To	Mode	Distance (km)
Shenzhen	Pune	Ocean Freight	4800
Stuttgart	Pune	Air Freight	6200
Ho Chi Minh City	Pune	Ocean Freight	3300
Pune	Mumbai	Heavy Truck	150

Exhibit 6: Component Weights

Component	Weight (kg)
Battery pack	22
Electric motor	18
Aluminum frame	26
Finished scooter (outbound)	85

Exhibit 7: Transport Emission Factors

Transport Mode	kg CO ₂ e / tonne-km
Ocean Freight	0.010
Air Freight	0.60
Heavy Truck (Diesel)	0.090

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
PhD/IDD Mid Semester Examination (2026)

Subject: **Foundations of Management**

Time: 2hrs

Part-A

Full Marks: 30

- The exam has two parts: Part(A)—20 Marks & Part(B)—10 Marks
- **Submit Part-B** along with the main answer script—failing which PART-B will not be evaluated
- Don't forget to write your Name and Roll number in part-B

Caselet 1: The Expansion Benchmark

FreshBrew Coffee, a fast-growing café chain, was evaluating expansion into Tier-2 Indian cities. During the first strategy meeting, an external consultant suggested that each new store should target monthly revenue of ₹18 lakh, citing performance of premium outlets in metro locations.

Although later internal feasibility studies projected that Tier-2 cities could realistically generate only ₹12–14 lakh, the original ₹18 lakh target became the reference point for discussions.

Regional managers began describing ₹14 lakh as “underperformance,” even though projections showed such stores would still be profitable. The finance team also rejected several viable locations because they fell short of the initial revenue benchmark.

By the time the final expansion plan was drafted, most decisions were still being evaluated relative to the consultant's first estimate rather than the updated market data.

- What bias is influencing decision making in this case?
- How did the initial number shape later judgments?
- What risks does this create for the expansion strategy?
- Suggest one managerial intervention to reduce this bias.

Caselet 2: The App That Had to Succeed

RetailKart, an online grocery startup, launched a mobile app with advanced personalization features. The CEO strongly believed the app would transform customer retention.

In the months after launch:

- The product team highlighted positive user reviews and rising download numbers.
- Marketing managers circulated testimonials from loyal customers.
- However, internal analytics showed that most new users stopped using the app within two weeks.
- Customer service reports also noted complaints about slow performance and confusing navigation.

Despite these warning signs, senior leaders focused mainly on the positive signals, continuing to invest heavily in promoting the app. Managers who questioned the strategy felt their concerns were being dismissed as overly pessimistic.

- Which behavioural bias is most evident in this case?
- Identify two examples where managers selectively interpreted information.
- How might this bias affect strategic outcomes for RetailKart?
- What process could the company introduce to ensure more balanced evaluation of evidence?

Caselet3: The Training Platform Decision at NexLearn

NexLearn, a mid-sized corporate training firm, had spent the past 18 months developing an in-house digital learning platform called NexLearn Pro. The system was designed to deliver customized leadership and technical training programs to corporate clients through an integrated app.

The company had already invested nearly ₹9 crore in software development, content production, and marketing preparation. Senior leadership frequently described the platform as “our future growth engine.”

However, during pilot testing, several issues emerged:

- clients preferred established third-party platforms they were already familiar with
- system maintenance costs were higher than expected
- the user interface required significant redesign

Rajiv Gandhi Institute of Petroleum Technology, Bangalore

B. Tech. End Semester Examination 2026

Subject: HU331 Organizational Psychology

Note: Attempt all questions

Time 3 hrs

Full Marks: 50

- ✓ Qu1. How do the Big Five personality traits influence workplace behavior and performance? Can certain traits predict success in specific job roles? Substantiate with examples. (10)
- ✓ Qu2. How has the digital age transformed traditional grapevine communication into online informal networks (e.g., WhatsApp, Slack, social media), and what new challenges does this pose? (10)
- ✓ Qu3. Please read the following case study and respond to the questions that follow.

"Remote Week" at HubSpot

When Rachel Leist was working in the marketing team at HubSpot, a technology company based in Cambridge, Massachusetts, the company noticed a change as it continued to grow. Many teams were now located in different parts of the world, which meant that meetings were often held online. Rachel's marketing team wanted to improve how they worked together and also help the company learn from their experience. As a result, they introduced a special initiative called "Remote Week."

During Remote Week, the team decided to conduct all their meetings through video calls. They experimented with different types of meetings, including one-to-one meetings, cross-functional meetings, and meetings involving the entire team. At the end of the week, feedback was collected from all team members to understand what worked well and what did not.

One important finding was that having a facilitator during video meetings made a significant difference. The facilitator ensured that everyone had an opportunity to speak. The team found that sometimes facilitators would see people making "goldfish faces" (opening and closing their mouths without saying anything as dominant viewpoints prevailed). The facilitator helped prevent this by managing turn-taking so that all voices were heard. Facilitators also paid attention to side conversations and encouraged participants to share those points with the entire group. Additionally, having everyone join the meeting from separate rooms helped reduce private side discussions.

The team also realized that informal, spontaneous conversations are very useful but difficult to achieve in virtual settings. To solve this, they created alternatives. Team members agreed to quickly join short video calls whenever someone had a question. If a person was unavailable,

Q7. An image is degraded by a linear, position-invariant system in the frequency domain, [10]
 modeled as: $G(u,v)=H(u,v)F(u,v)+N(u,v)$.
 The image undergoes the following degradations: Motion blur, Band-limited degradation,
 Periodic noise suppression using a notch filter and Additive Gaussian noise. Given the
 following degradation parameters:

- $H_1(u,v)=0.6$ (motion blur)
- $H_2(u,v)=0.5$ (band limitation)
- $H_3(u,v)=0.8$ (notch filter effect)
- Observed spectrum: $G(u,v)=200$
- Noise amplitude: $N(u,v)=30$
- Signal variance: $\sigma_f^2=400$
- Noise variance: $\sigma_n^2=36$

Perform the following Tasks:

- Compute the overall degradation function
- Restore the image using Wiener filtering
- Compare the result with inverse filtering and comment on stability of both the models in the presence of noise
- If the noise variance doubles, recompute the Wiener filter output and analyze how the restoration is affected.
- Compare the performance of both the above methods, if the degradation is instead caused by atmospheric turbulence, with parameters
 - $k=0.003$, $u=2$, $v=2$
 - $F(u,v)=150$, $N(u,v)=20$
 - Noise-to-signal power ratio $K=0.25$

Note: Use the following formulas wherever necessary and parameters can be assumed if needed

Let, $z=a+ib$, then $z^{-1} = \frac{a-ib}{a^2+b^2}$

Butterworth Low Pass Filter with order n : $H(u,v) = 1 / (1 + (D(u,v)/D_0)^{2n})$

Gaussian High-Pass Filter Equation: $H(u,v) = 1 - e^{-\frac{D^2(u,v)}{2D_0^2}}$

Spatial-domain Laplacian operator for image $f(x,y)$: $\nabla^2 f(x,y) = \partial^2 f / \partial x^2 + \partial^2 f / \partial y^2$

frequency-domain representation of the Laplacian operator: $\mathcal{F}\{\nabla^2 f(x,y)\} = -4\pi^2(u^2 + v^2)F(u,v)$

Adaptive noise reduction filter equation: $\hat{f}(x,y) = g(x,y) - \frac{\sigma_n^2}{\sigma_L^2(x,y)} [g(x,y) - m_L(x,y)]$

For atmospheric turbulence, $H(u,v) = e^{-k(u^2+v^2)^{5/6}}$

Q4. An image is corrupted by noise, and a neighborhood centered at pixel is given below. [5]
Assume that the original image values in the neighborhood are around 50–60, and the noise variance is $\sigma_n^2 = 25$.

$$\begin{bmatrix} 52 & 55 & 53 \\ 54 & 200 & 56 \\ 53 & 55 & 54 \end{bmatrix}$$

Perform the following tasks:

- Apply the adaptive local noise reduction filter to estimate the restored value at the center pixel.
- Apply the median filter to the same neighborhood and compute the restored pixel value.
- Compare the outputs of the two filters and explain which filter is more suitable in this case and why?

Q5. Describe each of the stages in the Canny Edge Detection algorithm with appropriate [10]
mathematical expressions wherever necessary. Given a 5×5 grayscale image:

$$I = \begin{bmatrix} 10 & 10 & 10 & 10 & 10 \\ 10 & 50 & 50 & 50 & 10 \\ 10 & 50 & 100 & 50 & 10 \\ 10 & 50 & 50 & 50 & 10 \\ 10 & 10 & 10 & 10 & 10 \end{bmatrix}$$

- Apply Canny Edge Detection with: Thresholds: Low = 30, High = 70 (assume already smooth for simplicity).
- Compare the above result using the thresholds: Low Threshold = 100, High Threshold = 210. Classify and show the pixels into strong, weak, and non-edge pixels.

Q6. Consider the following binary image A and two structuring elements are given: B1(Cross) [10]
and B2 (Square).

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$B_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Perform the below Tasks:

- Perform erosion of A using B1, followed by dilation of the result using B2.
- Perform dilation of A using B1, followed by erosion using B2.
- Compute the morphological gradient: $(A \oplus B_1) - (A \ominus B_1)$. Explain how the result relates to edge detection.
- Mention the key observations on Erosion, dilation, Opening, Closing, and Gradient, and Justify which operation is more suitable for removing small isolated noise in this image?

Rajiv Gandhi Institute Of Petroleum Technology, Jais, Amethi
IDD 4th Year
End Semester, January 2026- May 2026
Subject: CS431 Digital Image Processing

Time: 2.5 Hours

Full Marks: 50

INSTRUCTIONS:

1. Do not write anything other than your roll number on the question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Q1. A blur filter $h(m,n)$ has a 2D-DFT given by

[5]

$$H(u,v) = \begin{bmatrix} 1 & -0.3 - 0.3j & 0 & -0.3 + 0.3j \\ -0.3 - 0.3j & 0.1j & 0 & 0.1 \\ 0 & 0 & 0 & 0 \\ -0.3 + 0.3j & 0.1 & 0 & -0.1j \end{bmatrix}$$

Find the deblur filter $G(u,v)$ using the following approaches and also provide the significance in deblurring using such approaches

- 1) The inverse filtering approach
 - 2) The pseudo-inverse filtering approach, with $\delta = 0.05$
 - 3) The pseudo-inverse filtering approach, with $\delta = 0.2$
 - 4) Wiener filtering approach, with $\sigma_x^2 = 624$ and $\sigma_w^2 = 125$
- Compare all the above filtering approaches based on Numerical stability, Noise sensitivity and Restoration quality.

Q2. An 8×8 grayscale image $f(x,y)$ is transformed into the frequency domain as $F(u,v)$. At frequency coordinate $(u,v)=(6,7)$, the Fourier coefficient is: $F(u,v)=6+8i$. The cutoff frequency is $D_0 = 3$, and the center of the frequency plane is $(4,4)$. Considering that a Butterworth Low Pass Filter (order $n=2$) is applied, perform the following tasks:

- A. Evaluate the filter response $H(u,v)$ and filter output at $(6,7)$.
- B. If instead a Gaussian High Pass Filter were applied, how the filter output would change.
- C. Comment on how both the filter affects on Edge enhancement and Noise amplification

Q3. An image $f(x,y)$ is transformed into the frequency domain as $F(u,v)$. [5]

- A. Using the Fourier Transform properties of derivatives, derive the frequency-domain representation of the Laplacian operator.
- B. For a frequency component at $(u,v)=(2,3)$, the Fourier transform value is $F(u,v)=5$. Compute the filtered output using the laplacian transfer function in the frequency domain.
- C. Why is the Laplacian typically subtracted from the original image instead of added?

Arusha IT 20

they recorded a short video message so that communication was not delayed and no one was excluded.

Another key learning was the importance of staying visible when working remotely. Team members made an effort to share their ideas during virtual meetings. HubSpot also used Slack to communicate across teams. By participating in Slack discussions, whether about work tasks or casual topics, employees remained connected and visible to colleagues working in different locations.

According to Rachel, Remote Week was a success. The team improved collaboration across locations and felt more confident that everyone on the team had an opportunity to be heard.

Questions:

- ✓ (a) Using Solomon Asch's experiment, analyze why some team members may hesitate to speak up in virtual meetings. How did HubSpot's "Remote Week" practices help address this issue? (10)
- ✓ (b) Which leadership style is most evident in the HubSpot "Remote Week" initiative? Justify your answer with examples from the case. (10)
- ✓ (c) How can the Johari Window be used to improve openness and trust in virtual teams? (10)

Rajiv Gandhi Institute of Petroleum Technology, Bangalore
B. Tech. IDD

Mid-Semester Examination 2026
Subject: HU331 Organizational Psychology
Note: Attempt all questions

Time 2 hrs

Full Marks: 30

1. Critically evaluate the limitations of bureaucratic organizational structures in light of contemporary organizational changes. (7.5)
2. Among major motivation theories, which offers the strongest explanatory and practical value in contemporary organizational settings? Justify your answer critically. (7.5)
3. How do early childhood experiences shape the strength and balance of id, ego, and superego? Explain with suitable examples. (7.5)
4. If you were a recruiter for a customer-service call centre, what personality types you would prefer to hire and why? How might emotional intelligence play a role for call centre employees in responding to abusive customers? Should companies allow a certain degree of abuse if that abuse results in satisfied customers and perhaps greater profit? What are the ethical implications of this? (7.5)

- c. (2 Marks) Explain how the outputs from different heads are merged to form the final multi-head attention representation.
- d. (6 Marks) Consider a multi-head attention setup as given below with assuming 2 tokens and 2 heads, with key/query dimension, $d_k = 2$, value dimension, $d_v = 2$. For each head $i \in \{1, 2\}$, you are given the following head details:

Head 1:

$$Q^{(1)} = K^{(1)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad V^{(1)} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Head 2:

$$Q^{(2)} = K^{(2)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad V^{(2)} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

Calculate the following:

- i. The scaled scores, $S^{(i)}$
- ii. Attention weights, $A^{(i)}$
- iii. The final multi-head output, $O^{(i)}$

[Formula: softmax equation = $\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^n \exp(z_j)}$]

SECTION B (25 Marks)

- ❖ Each question in SECTION B carries 1 mark, for a total of 25 marks across 25 MCQs.
- ❖ Answer by writing only the correct option (A, B, C, or D) for each question in order

Q1. In attention mechanisms, Key, and Value represents?

- A) Key: What a word is looking for in other words, Value: The information to be retrieved after attention
- B) Key: The information to be retrieved, Value: What each word offers to match against queries
- C) Key: What each word offers to match against queries, Value: The information to be retrieved
- D) Key: The meaning of the word, Value: The sum of embeddings

Q2. Which set correctly corresponds to the shapes of Query (Q), Key (K), and Value (V) for a sequence length n and query/key depth d_k , and value depth d_v ?

- A) Q: $n \times d_k$, K: $n \times d_k$, V: $n \times d_v$
- B) Q: $n \times d_k$, K: $d_k \times d_k$, V: $d_v \times n$
- C) Q: $n \times d_k$, K: $d_k \times n$, V: $d_v \times n$
- D) All must be $n \times n$

Q3. Transformers remove which of the following components entirely?

- A) Convolution layers
- B) Positional embeddings
- C) RNN recurrence
- D) Feed-forward layers

Q4. In a Transformer with 6 encoder layers and 6 decoder layers, how many total multi-head attention modules are used?

- A) 6 in encoder; 6 in decoder
- B) 12 in encoder; 12 in decoder
- C) 6 in encoder; 12 in decoder
- D) Only 6 in encoder

Q5. In a standard scaled dot-product attention, a single head with sequence length $T = 2048$ and feature dimension $d = 64$, approximately how many scalar multiplications are required for computing QK^T ?

- A) $2048^2 \times 64$
- B) 2048×64
- C) 64×64
- D) 2048×2048

Q6. For an odd dimension index $2k + 1$, the positional encoding component is:

- A) $\sin(\text{pos} \cdot \omega_k)$
- B) $\cos(\text{pos} \cdot \omega_k)$
- C) $\tan(\text{pos} \cdot \omega_k)$
- D) $\sin(\text{pos} \cdot k)$

Time 3 hrs

Full Marks: 60

1. Drawing on multiple theoretical perspectives and empirical studies covered in class, analyze the key factors that shape consumer purchase decisions for green, energy-saving gadgets. (20)
2. Identify the key obstacles hindering India's energy transition and analyze how an individual's habitus reinforces or reproduces the barriers. (10)
3. How can media stories help construct identities such as the "eco-friendly citizen," "modern energy consumer," or "nationalist supporter of energy independence"? What implications does this have for public attitudes and energy behaviours? Explain with suitable examples. (10)
4. A solar-module manufacturer sources aluminium frames from the following two suppliers:

	Supplier A	Supplier B
Base Price (INR/unit)	1,000	1,150
Emission Factor (kg CO ₂ /unit)	45	20

The firm applies an internal carbon price of Rs. 4,000 per tCO₂.

- (a) Compute the internal carbon cost for each supplier. Based on these values, determine the carbon-adjusted procurement cost per unit. (10)
- (b) Determine the break-even internal carbon price p (INR per tCO₂) at which Supplier B becomes financially preferable to Supplier A. (10)

	(c) Panic and leave the office (d) Blame the testing team without checking anything	
10.	Which LinkedIn "About" summary is most aligned with employability and industry readiness for a final-year engineering student? (a) "Just here for fun. Add me if you want memes." (b) "B. Tech (EEE) student with internship experience in energy analytics; interested in EV systems, MATLAB/Python, and data-driven problem solving. Open to roles in EV, energy analytics, and microgrid projects." (c) "I need a job urgently in any field." (d) "Proud to be from XYZ college. Contact for all kinds of work."	1

SECTION B

Attempt **ALL** Parts and Answer in short.

(5x4)

1.	Define "employability skills". List any four key employability skills that are important for engineering graduates.	4
2.	What are the main components of an effective résumé for fresh graduates? Mention at least five key components.	4
3.	Write a short note on GD (Group Discussion): purpose, typical evaluation parameters, and two common mistakes made by students.	4
4.	Explain time management techniques that a final-year student can use to balance academics, placement preparation, and projects.	4
5.	What do you understand by industry readiness? Highlight any four gaps commonly observed between academia and industry expectations.	4

SECTION C

Attempt **ALL** Parts

(2x10)

1.	(a) What is a career development roadmap? Design a 3-year career development plan from 2 nd year to 4 th year for a B. Tech graduate focusing on skills, certifications, projects, and roles. (b) How will this roadmap help in Industry readiness, long-term employability, growth, and adaptability in a rapidly changing industry?	5+5
2.	(a) Explain the complete process of preparing for a job interview – from researching the company to post-interview follow-up. (b) Draft a sample set of 5 HR interview questions and provide ideal, concise answers from the point of view of a B. Tech final-year student.	5+5

Q14. BERT is mainly used for:

- A) Text generation, because BERT's attention enables predicting the next token sequentially.
- B) Text understanding, because BERT learns contextual bidirectional representations
- C) Image recognition, because BERT's transformer encoder is optimized for spatial feature extraction.
- D) Speech synthesis, because BERT's masked language modeling maps text to audio embeddings.

Q15. GPT's training objective minimizes:

- A) $\sum_t -\log(P(x_t | x_{<t}))$
- B) $\sum_t -\log(P(x_t | x_{>t}))$
- C) $\sum_t -\log(P(x_t | \text{masked input}))$
- D) Mean squared error between token embeddings

Q16. CLIP is trained to:

- A) Predict the next word in a sentence
- B) Reconstruct masked tokens in text
- C) Align image and text embeddings
- D) Generate images from text using a diffusion objective

Q17. CLIP uses which training objective?

- A) Cross-entropy loss
- B) Contrastive loss
- C) Reconstruction loss
- D) GAN adversarial loss

Q18. DALL·E is primarily a model for:

- A) Bidirectional text understanding using masked token prediction
- B) Autoregressive image generation conditioned on textual prompts
- C) Contrastive alignment of image-text pairs to learn shared embeddings
- D) Two-stage encoder-decoder summarization

Q19. DALL·E is capable of:

- A) Code generation
- B) Speech recognition
- C) Image generation from text prompts
- D) Machine translation

Q20. Compared to standard Mixture-of-Experts (MoE), Mixture-of-Recursions (MoR) offers:

- A) Less flexibility in modeling long-range structure
- B) Improved hierarchical reasoning
- C) Slower inference due to activating all experts
- D) Fewer specialized experts and reduced modularity

Q7. In the scaled dot-product attention, if the key dimension is $d=64$, what is the scaling factor?

- A) $1/4$
- B) $1/8$
- C) $1/16$
- D) $1/256$

Q8. How many attention score computations are required if computing with standard attention for sequence length $T=3000$, vs sparse attention if each token attends to only 100 neighbors?

- A) Standard: 9,000,000 | Sparse: 300,000
- B) Standard: 3000 | Sparse: 100
- C) Standard: 3,000,000 | Sparse: 300,000
- D) Standard: 9,000,000 | Sparse: 900,000

Q9. For sinusoidal positional encoding, which option correctly gives the formulas for the even and odd dimension components?

- A) Even index $2k$: $\cos(\text{pos} \cdot \omega_k)$, Odd index $2k+1$: $\sin(\text{pos} \cdot \omega_k)$
- B) Even index $2k$: $\sin(\text{pos} \cdot \omega_k)$, Odd index $2k+1$: $\cos(\text{pos} \cdot \omega_k)$
- C) Even index $2k$: $\text{pos} \cdot \omega_k$, Odd index $2k+1$: pos/ω_k
- D) Even index $2k$: 0, Odd index $2k+1$: 1

Q10. If a ViT processes an image into m patches each of dimension d , and VPT-shallow adds p prompt vectors, the input to the transformer becomes:

- A. $m \times d$
- B. $(m+p) \times d$
- C. $(m \cdot p) \times d$
- D. $m \times (d+p)$

Q11. BERT uses which objective during pre-training?

- A) Next-token prediction
- B) Next-sentence prediction
- C) Masked-token prediction
- D) Sequence classification objective

Q12. GPT, BERT and DALL-E use which part of the Transformer architecture?

- A) Encoder only
- B) Decoder, Decoder and Encoder respectively
- C) Both encoder and decoder in all three
- D) Encoder, Decoder and Decoder respectively

Q13. What is the key difference between GPT and BERT?

- A) GPT uses bidirectional attention, BERT uses unidirectional attention
- B) BERT uses autoregressive decoding
- C) BERT uses bidirectional attention, GPT uses unidirectional attention
- D) Both use the same attention mechanism

- projected revenue was revised downward by almost 40 percent

The strategy team presented two options to the executive committee:

Option A: Continue investing another ₹4 crore to upgrade the platform and push for market adoption.

Option B: Stop further development, license an existing external platform, and focus resources on content creation instead.

During the discussion, opinions divided sharply.

Some executives argued that abandoning NexLearn Pro would mean wasting the ₹9 crore already spent. Others felt the company had publicly positioned the platform as a strategic innovation and withdrawing now might signal failure. The CFO, however, pointed out that the earlier investment could not be recovered regardless of the next decision. What mattered, she said, was which option would produce better future returns.

As the meeting ended, the CEO asked the team to prepare a final recommendation by the following week.

- a) If you were the CEO, what framework would you use to ensure the decision focuses on future costs and benefits rather than past spending?
- b) Suggest one organizational practice that could help managers avoid this type of decision trap in future projects.

Caselet-4: NexaCloud Technologies cloud contracts

NexaCloud Technologies is a fast-growing IT services firm providing cloud migration and cybersecurity solutions to global clients. A key banking client has issued a sudden Request for Proposal (RFP) for a large-scale cloud modernization project, with a submission deadline of 72 hours. Winning the contract would increase NexaCloud's annual revenues by nearly 25%.

The proposal team presents four sharply different recommendations:

- The Analytics Team uses historical bid data, pricing models, and win-loss statistics to argue that NexaCloud should submit a standardized proposal at a conservative price point, which historically maximizes win probability.
- The Delivery Head argues that the historical data is only partially relevant because the client's current RFP includes new regulatory and cybersecurity requirements not present in earlier projects.
- The CEO, who has personally handled the client relationship for over a decade, insists—based on her “feel for the client”—that an aggressive, customized proposal with a premium price will be perceived as confidence and capability.
- The HR Head cautions that the firm is already operating at near-full utilization and that absorbing the project may overstretch teams, increasing attrition risk.

Available data is incomplete, some of it contradictory, and the cost of a wrong decision is very high. The leadership team must decide whether to bid, and if so, how aggressively.

a) Identify which recommendation best reflects each of the following decision-making approaches:

- i) Rational decision making; ii) Bounded rationality iii) Intuitive decision making iv) Evidence-based management

b) Explain why pure rational decision making is difficult to apply in this situation, despite the availability of analytical tools.

c) Critically evaluate the risks of over-reliance on intuition in this case. Identify two specific bias risks

Rajiv Gandhi Institute of Petroleum Technology, Bengaluru
B. Tech./IDD End Semester Examination (Odd Sem, 2025-26)

Subject: Principles of Economics

Time: 3 hrs

PART-A

Full Marks: 60

The exam has two parts: Part A with full-length questions (40 marks) and Part B with MCQs (20 marks). Answer all the questions. Use diagrams wherever possible to support and explain your answers clearly.
PART-A: Questions Q1–Q5 carry 5 marks each, while Q6 and Q7 carry 7.5 marks each

Q1. The government proposes a *Luxury Tech Tax* on premium iPhones and Apple Watches, aiming to collect revenue from affluent consumers. In the short run, demand for high-end Apple devices is relatively elastic (buyers can delay upgrades or choose alternatives), while Apple's supply is relatively inelastic (production plans are fixed).
(a) Based on elasticity and tax incidence, explain who will bear most of the tax burden.
(b) Illustrate your answer with a tax-incidence diagram using an elastic demand curve and an inelastic supply curve.

Q2. Digital platforms like food-delivery apps or ride-sharing apps match thousands of buyers and sellers every minute. Using the idea of **Adam Smith's Invisible Hand**, explain how such platforms achieve efficient outcomes without any central planner deciding prices or quantities.

In your answer, comment on:

- i. How individual decisions by buyers and sellers generate market-clearing prices,
- ii. Why the resulting allocation is economically efficient, and
- iii. How this illustrates the idea that free markets are often the best way to organize economic activity.

Q3. A technology company develops a **new graphene-based ultra-fast charging chip** for electric vehicles (EVs). Because the firm holds an exclusive patent, it is the **only producer** of this chip for several years. EV manufacturers must purchase the chip from this company if they want to offer fast-charging features in their vehicles. After the patent expires, several electronics firms start producing **identical versions** of the same chip. Prices in the market begin to fall, and EV manufacturers now have many suppliers to choose from.

- i. **Market Structure Identification:**
Explain why the chip market behaves like a **monopoly** during the patent period and like **perfect competition** after the patent expiry. Refer to entry barriers, number of firms, and pricing power in your explanation.
- ii. **Engineering & Consumer Impact:**
Describe how the **price, availability, and adoption** of the fast-charging chip are likely to differ under monopoly vs perfect competition.
 - a. How does a monopoly affect EV manufacturers and EV buyers?
 - b. How does competition change incentives for innovation and cost reduction?

Q4. FreshFizz Beverages and CoolQuench Drinks are the two major sellers of 500 ml soda bottles in a city. For years, their prices have stayed around ₹40. FreshFizz observes that:

- If it **raises** its price, CoolQuench does **not** match → FreshFizz loses many customers (demand is **elastic** for price increases).
- If it **cuts** its price, CoolQuench matches immediately → FreshFizz gains very few customers (demand is **inelastic** for price cuts).

FreshFizz concludes that it faces a **kinked demand curve** at the current price.

Tasks

(a) Draw a **kinked demand curve** and the corresponding **discontinuous MR curve**, labelling the elastic and inelastic segments.

(b) Explain briefly **why prices remain rigid** in this duopoly.

(c) If FreshFizz's marginal cost rises slightly (e.g., ₹20 → ₹22), use your diagram to explain whether the firm will **change its price**.

(d) Give **one real-world industry** where the kinked demand curve helps explain stable prices.

Q5. Two competing smartphone brands — **TechNova** and **MobiQuest** — are preparing to launch mid-range models in the Indian market. Each firm must choose between two pricing strategies:

- **H:** High Price (premium positioning)
- **L:** Low Price (aggressive penetration pricing)



Rajiv Gandhi Institute of Petroleum Technology
Bengaluru Campus
End-Semester Examination
(ODD Semester, 2025-26)

Max. Marks: 60 **Duration:** 03 Hrs

Date: 04/12/2025 **Roll No:** 22 CS 2002

Programme/Semester: B. Tech 7th Semester

Time: 03:00 PM-06:00 PM

Course Title: Employability Skills and Industry Readiness

S.NO	Questions	Marks
SECTION A (Objectives)		
Attempt ALL Parts and select the best option		(10x1)
1.	A company is hiring a "Graduate Trainee – Data & Operations." The JD stresses: Excel/SQL skills, stakeholder communication, and willingness to work in shifts. Which summary line on your CV best shows alignment with this role? (a) B.Tech student interested in music, sports, and travelling (b) Final-year B.Tech student with course projects in data analysis using Excel/SQL and experience coordinating department events (c) Fresher looking for any job in any domain (d) B.Tech student who prefers only research and higher studies	1
2.	You are sending a follow-up email to an HR after completing two interview rounds. What is the most appropriate subject line? (a) URGENT!!! When will you give me the job??? (b) Regarding my salary and promotion (c) Follow-up on Interview for Graduate Engineer Trainee – [Your Name] (d) Hi, remember me?	1
3.	An online test has 40 questions, +1 mark for every correct answer and –0.25 for every wrong answer. You are unsure about 10 questions. What is the most employable and exam-smart strategy? (a) Attempt all 40 questions randomly (b) Attempt only the questions you are reasonably confident about and make educated guesses for a few borderline ones (c) Attempt only 10 questions and submit the test early (d) Leave the test midway when you see difficult questions	1
4.	You are the team coordinator for a final-year project. One member is continuously missing meetings and deadlines. What is the most professional first step? (a) Complain directly to the Head of Department without informing the member	1

- A. $P(\omega_1) > P(\omega_2)$
- B. $p(x|\omega_1)/p(x|\omega_2) > P(\omega_2)/P(\omega_1)$
- C. $p(x|\omega_1)P(\omega_1) < p(x|\omega_2)P(\omega_2)$
- D. $P(\omega_1|x) < P(\omega_2|x)$

24. In a two-class problem, suppose: $p(x|\omega_1)=0.3$, $p(x|\omega_2)=0.2$, $P(\omega_1)=0.4$, $P(\omega_2)=0.6$. Using Bayes' rule, which class should we assign x to?

- A. ω_1
- B. ω_2
- C. Equal likelihood
- D. Cannot decide

Q25. The zero-one loss function assumes that:

- A. All errors have equal cost
- B. Errors have varying costs depending on the class
- C. Misclassification is ignored
- D. Loss depends only on feature x

Q26. In K-Means clustering algorithm, given the two points A(2, 3) and B(8, 9) assigned to Centroid C1 at (5, 6), compute the squared Euclidean distance sum for C1.

- A. 13
- B. 18
- C. 25
- D. 36

Q27. Which clustering algorithm does not require the number of clusters (k) to be specified in advance?

- A. K-Means
- B. DBSCAN
- C. K-Medoids
- D. Partitioning Around Medoids (PAM)

Q28. In density-based clustering like DBSCAN, the two key parameters are:

- A. k and max-Iterations
- B. ϵ (epsilon) and MinPts
- C. Number of clusters and distance metric
- D. Centroid and variance

Q29. A point cloud is typically:

- A) structured data
- B) color-coded pixels
- C) unstructured 3D point set
- D) 2D histogram

Q30. If a voxel is too large relative to the object size, the resulting 3D shape becomes:

- A) highly precise
- B) blurred and blocky
- C) transparent
- D) invisible